Protocol choice changes the winner more than the ranking: a factorial audit of five negative-set rules and ten peptide bioactivity tasks from a single curated collection

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Abstract

Background. Homology-aware data splitting is now standard in peptide property prediction, and recent benchmarks enforce it by construction. Whether that correction makes results comparable, and whether the splitting method delivers what it is assumed to, have not been measured.

Methods. We held 16–24 model pipelines fixed across two crossed designs. Arm B fixes the task and varies the protocol: four clustering thresholds × five negative-set rules × five seeds, on one harmonised antibacterial corpus (5,187 published positives; negatives drawn from reviewed UniProt under each rule). Arm A reverses this, fixing the protocol and varying the biological task across ten datasets (52,316 sequences as released, 52,001 after removing 315 sequences duplicated across their own train/test split) drawn from **a single collection curated by one group with one homology-aware partitioning pipeline**, under their native partitions. Residual leakage and class balance were measured per cell rather than assumed; thresholds and a minimum detectable effect were fixed in a dated analysis plan, released with every deviation logged. Rank agreement and winner stability are each judged against a *measured* reproducibility ceiling from seed replicates rather than a nominal maximum.

Results. The negative-set rule accounted for 71.0% of the variance in reported MCC against 14.8% for the pipeline: which negatives a benchmark invents matters more to its headline number than which model is run on it, and mean MCC spanned 0.31 to 0.91 across protocol cells. **Our primary hypothesis – that leakage-controlled protocols rank methods differently – was not supported**: normalised agreement across the five negative-set rules was 0.881 against a pre-specified threshold of 0.85 ($\tau_b = 0.727$). Broad ordering was comparatively stable while the identity of the top-ranked pipeline was not, and the two must be read separately. Rankings transferred less well across the ten tasks than across the five rules (normalised 0.572; difference +0.218, $p = 0.024$ when the comparison is generalised over levels rather than over pipelines). Winner identity was the fragile quantity, but it is noisy in itself: reseeding alone changed the top-ranked pipeline in 54.5% of comparisons, and against that ceiling the negative rule reached 76.0% and the task 93.3%. Only the task figure survives resampling of the levels; with five rules the protocol flip rate is a measurement on those rules, not an inference about protocols. A two-feature length-and-charge baseline reached MCC 0.79 on proteome-derived negatives, within 0.11 of a 650M-parameter language model, and fell to chance once that shortcut was removed. Cluster-disjoint splitting itself underdelivered: residual pairwise leakage rose to 21.6% of test sequences as the nominal threshold *tightened*, test class balance reached 99.5% positive on a 1:1 corpus, the split inverted the sign of a discriminative feature in two tasks, and 11 of 16 published datasets contained exact duplicates across their own partition.

Scope. These are conclusions about *five negative-set rules applied to one fixed positive set* and *ten bioactivity tasks curated by one group under one partitioning pipeline*. They are not evidence that independently constructed benchmarks disagree in general: that comparison is not performed here and remains open. Intervals that generalise beyond the specific levels studied are wide, and we report them as such.
Conclusions. Leakage control did not make evaluation interchangeable across these levels, but the failure is specific: protocol choice disturbs the broad ordering modestly while changing which method is crowned far more often – and single-run benchmarks cannot identify a winner reliably even against a rerun of themselves. We give reporting requirements, introduce a balance-preserving cluster-disjoint splitter as a clearly labelled post hoc sensitivity analysis, and release the harness and all 4,400 run records.

Keywords: benchmark reliability; antimicrobial peptides; data leakage; evaluation protocol; homology-aware splitting; rank agreement; protein language models

1 Introduction

Computational prediction of peptide properties, and of antimicrobial activity in particular, has an unusual recent history. For most of the last decade, published classifiers reported accuracies that practitioners could not reproduce on new sequences. The diagnosis, now widely accepted, was data leakage: biological sequence databases are redundant by construction, so a random train–test split places homologues on both sides of the partition and the model is rewarded for recognising families rather than for learning the property.

The field responded. Homology-aware partitioning is now the default rather than the exception, and a cluster of recent resources enforces it by construction [1–6]. Reported performance dropped, as expected, and the drop was interpreted as the removal of an artefact. A reasonable conclusion would be that the problem has been solved.

This paper tests a narrower and more awkward question. Leakage control fixes the *level* of reported performance. It does not obviously fix the *ordering* of methods, and ordering is what the literature actually trades in: a new method is adopted because it placed first on some benchmark. If evaluations that are equally leakage-controlled rank the same methods differently, then a claim of state of the art is a statement about an evaluation protocol rather than about biology.

There is reason to expect exactly this. Leakage control is not a single decision but a family of them: a similarity measure, a threshold, a clustering tool, a coverage parameter, a partitioning algorithm. More consequentially for antimicrobial-peptide prediction, a benchmark must *invent a negative class*. There is no natural population of non-antimicrobial peptides, so every benchmark constructs one, and the construction determines the difficulty, the apparent winner, and what the model learns. Sidorczuk et al. [1] held the positive set fixed, varied only the negative-sampling method across eleven published definitions, and concluded that the field did not know which predictor was best. That study predates routine homology control; whether its conclusion survives is unknown. The need for a common evaluation standard has since been argued normatively [7], and related work shows that leakage can persist even under similarity-based splitting in protein interaction benchmarks [8] and that performance estimates are properties of whole data-centric workflows rather than of models [9].

Two recent single-task studies suggest it may. In peptide–protein affinity prediction the identity of the held-out entity changes not only the score but the winner [10]; in kinase bioactivity prediction a random forest on Morgan fingerprints matches or beats every neural architecture once scaffold or target-held-out splitting is used [11]. Both note, and neither resolves, that systematic comparisons across splitting strategies on matched data remain uncommon.

Meanwhile the machine-learning community has developed a quantitative apparatus for exactly this question. Work on language-model leaderboards shows rankings moving by many positions under perturbations practitioners regard as implementation details [12], and formalises

multi-task benchmarking as a social choice problem in which diversity and stability trade off against one another [13]. Importing that apparatus is part of what we contribute.

Table 1: Position relative to the closest prior work. *Ceiling* means the agreement statistic is compared against a measured reproducibility floor rather than its nominal maximum; *pre-reg.* means decision thresholds were fixed before analysis.

Study	Domain	Varied	Measured	Ceiling	Pre-reg.
Sidorczuk et al. [1]	AMP	negative rule (11)	scores, no homol. ctrl.	–	–
Tian et al. [10]	pep.–prot.	held-out entity	score + winner	–	–
Abbott [11]	kinase	split strategy	score + winner	–	–
Bushuiev et al. [8]	PPI	similarity split	residual leakage	–	–
Alzahrani et al. [12]	LLM	impl. detail	rank movement	–	–
Zhang and Hardt [13]	LLM	benchmark choice	rank agreement	✓	–
Cisterna-García et al. [7]	AMP	– (normative)	–	–	–
This work	peptide	**protocol *and* task**	rank, winner, leakage	✓	✓

Contributions

We hold a fixed set of pipelines constant across **two complementary crossed designs**. Arm B holds the task fixed and varies the protocol; Arm A holds the protocol fixed and varies the biological task. Together they separate the two sources of ranking movement; we are not aware of a study that crosses both in one design, though each has been varied separately. Decision thresholds, including a minimum detectable effect, were fixed in a dated analysis plan before any data were analysed; the plan was not externally deposited, and Section 2.6 lists every deviation from it, including one that changes what H1 was tested over.

1. **The negative-set rule, not the model, sets the reported number.** It accounts for 71.0% of the variance in MCC against 14.8% for the pipeline, from exact method-of-moments variance components on a complete balanced factorial. All protocol-attributable *reordering* is carried by the pipeline-by-protocol interaction, in which the negative rule outweighs the clustering threshold sevenfold.

2. **Task transfer is the larger problem.** Rank agreement is lower across the ten tasks than across the five rules: $+0.218$, $p = 0.024$ generalising over levels, under the pre-registered splitter. The same contrast is $+0.185$, $p = 0.141$ over levels under the post hoc splitter, so the claim rests on the pre-registered analysis; both are nominally significant if the levels are instead held fixed and only pipelines resampled ($p = 0.001$ and $p < 0.001$), which answers a narrower question (Section 5.5). The field has a convention for neither factor.

3. **Both agreement statistics have measurable ceilings below their nominal maxima**, obtainable free from seed replicates, and we report both against them. The ceiling on winner identity is the more alarming: seed noise alone changes the top-ranked pipeline in 54.5% of comparisons, so a single-run leaderboard cannot identify a best method reliably even against a rerun of itself. Across tasks the excess over that ceiling survives generalisation over levels; across the five negative-set rules it does not, and we keep the latter descriptive.

4. **Cluster-disjoint splitting does not deliver what it is assumed to**, in four separate ways that we measure rather than assert.

5. **A trivial baseline diagnoses shortcut-solvable benchmarks**, and a stress test of four cluster-disjoint splitters locates the cause of two artefacts in *size-ordered* cluster assignment: shuffling cluster order halves residual leakage and nearly fixes class balance at once, where our own quota-driven splitter fixes balance exactly but leaks more than the method it replaced (Section 2.5).

Our primary hypothesis was *not* supported, and we report it as such: the headline rank-agreement result is a null against a threshold we had fixed in advance and did not move. What makes the rest of the paper credible is that threshold, together with the deviations log that shows where the executed design departed from the plan.

2 Methods

2.1 Two crossed designs

Arm B (vary the protocol). One harmonised antibacterial corpus; the evaluation protocol is crossed over four clustering thresholds (random, 90%, 70%, 40% identity), five negative-set rules, five seeds and six metrics. $4 \times 5 \times 5 = 100$ cells, each scored by 16 pipelines: 1,600 fits. The grid was run twice, once with the splitter as pre-registered and once with the balance-preserving splitter of Section 2.5: 3,200 fits in total.

Arm A (vary the task). Ten curated peptide bioactivity datasets, each under its own homology-aware partition as released, scored by 24 pipelines with five seeds: 1,200 fits. Tasks: antibacterial (13,245/3,311 train/test), antimicrobial (10,266/2,566), antiviral (4,711/1,177), neuropeptide (3,881/969), toxicity (3,092/772), ACE-inhibitory (1,685/421), antifungal (1,591/395), anticancer (1,378/344), DPP-IV (1,063/265) and tumour T-cell antigen (948/236); all balanced near 1:1 and fully canonical.

Scope of Arm A. The ten datasets were curated by one group with one pipeline [2], so Arm A asks whether a ranking established on one bioactivity transfers to another *at a fixed protocol*. What this does and does not license is set out once, in Section 6.1, and not repeated here.

Why ten of the sixteen released datasets. The collection contains sixteen. We ranked pipelines on the ten with at least 900 training sequences, because a 24-pipeline leaderboard built on a few hundred sequences is dominated by test-set noise. The cutoff falls in a natural gap: the ten retained sets have 948–13,245 training sequences and the six excluded ones 191–698. Even so the retained small tasks are near the edge – the measured seed ceiling is 0.826 for tumour T-cell antigen ($n_{\text{train}} = 948$) and 0.754 for DPP-IV (1,063), against 0.909 for antimicrobial (10,266) – which is why every agreement statistic in Arm A is reported against that measured ceiling rather than against 1.0. The cutoff was fixed on dataset size alone, before any pipeline was fitted and therefore before any ranking existed to inspect; it is recorded in the frozen task list in the released harness. The duplicate audit in Section 4.4(d) covers **all sixteen**, since it needs no model fits. We report that audit over the full collection and the ranking analysis over the ten, and flag the difference wherever a count appears.

2.2 Corpus construction for Arm B

Positives. The published positive set of Sidorczuk et al. [1]: DBAASP records active against Gram-positive or Gram-negative bacteria, non-standard residues removed, length ≥ 5, reduced at 90% identity with CD-HIT. We removed three further sequences (`PPPPPPPPPP`, `DDDDDDD`, `FFFFFFFF`) because no permutation of a homopolymer differs from the sequence itself, so under the shuffled-positive rule they would be emitted as their own negatives. They were excluded under *every* rule, so the positive set is identical across rules. Final: **5,187 sequences**.

Negative pool. Reviewed UniProt [14], retrieved 2026-10-05: 575,749 records, of which 573,063 contain only the 20 standard residues. Duplicate *sequences* were removed, since SwissProt stores identical sequences under different accessions and sampling rows without replacement can otherwise draw one sequence twice.

Negative-set rules. Five rules spanning the families behind the eleven methods catalogued by Sidorczuk et al. [1], whose definitions we read from their published source: (i) *random UniProt*,

uniform draw length-filtered to 5–100; (ii) *annotation filtered*, as (i) but excluding records annotated *Antimicrobial*, *Antibiotic*, *Fungicide* or *Defensin*; (iii) *length matched*, random contiguous fragments matching the positive length distribution; (iv) *composition matched*, fragments chosen from 60 candidates per positive to minimise L_1 distance in amino-acid composition; (v) *shuffled positives*, each positive randomly permuted.

Class ratio was fixed at 1:1 for every rule, although published methods use ratios from 1:1 to 1:10, because F1 and AUPRC respond to class balance and letting the ratio travel with the rule would confound the two. No negative is byte-identical to any positive, asserted at construction. Each corpus contains 10,374 sequences.

2.3 Model pipelines

A pipeline is a representation crossed with a learner; the factorisation is what allows representation and learner rank stability to be compared by marginalising over one axis.

Representations (9): amino-acid composition; AAC + dipeptide composition; a physicochemical vector (net charge, mean Kyte–Doolittle hydropathy, hydrophobic moment at 100° periodicity, aromaticity, aliphatic index, length, charge density); character k-mer TF–IDF ($k = 1$–3, fitted on the training partition only); mean-pooled embeddings from **ESM-2** [15] at four scales (8M, 35M, 150M, 650M); and a deliberate two-feature null of sequence length and net charge.

Learners (5): L_2 logistic regression; random forest (300 trees); an approximate RBF support vector machine (Nyström map, 512 components [16], followed by a linear SVM with sigmoid-calibrated probabilities); a single-hidden-layer MLP (256 units); gradient-boosted trees [17]. Features are standardised in every pipeline, including tree-based ones, so pipelines are not quietly differentiated by preprocessing. Models were fitted with scikit-learn [18]; language-model embeddings used PyTorch [19] and the Transformers library [20]; statistical routines used SciPy [21].

Arm B used 16 pipelines; Arm A used 24, the eight additional ones being new learners on representations already cached, which complete the representation × learner grid at no extra embedding cost. **Hyperparameters were fixed before the corpus was built and were never tuned per cell**, so that protocol preference cannot be confounded with search effort. Section 3.7 tests what that choice cost by giving every pipeline an equal tuning budget under group-aware validation.

Exclusions. End-to-end convolutional and recurrent sequence models were excluded because, unlike every other pipeline, their cost cannot be amortised by caching features, and the available hardware was CPU-only. ProtT5-XL (3B parameters) was excluded as CPU-infeasible. Both are logged as pre-registration deviations.

2.4 Splitting, and what was measured rather than assumed

Within each Arm B cell, sequences were clustered with MMseqs2 [22] and whole clusters assigned to a held-out 20% partition. The call was `easy-cluster` with `--min-seq-id` set to the cell threshold, coverage `-c 0.8`, `--cov-mode 0` and sensitivity `-s 7.5`. We report the coverage parameter because it is routinely omitted although for short peptides it changes cluster structure more than the identity threshold does.

Leakage was measured, not asserted. An assertion that no train–test pair exceeded the threshold failed on real data. Greedy clustering does not form equivalence classes: two sequences can occupy different clusters and still exceed the threshold, so cluster-disjoint does not imply pairwise-disjoint. Every cell therefore records the fraction of test sequences with an above-threshold hit in the training partition, found by explicit MMseqs2 search. Exact duplicates remain fatal.

Fairness invariants, asserted per cell. (i) All pipelines receive the identical training index and label vector, checked by hash. (ii) Transforms that learn anything – the TF–IDF vocabulary, the standardiser statistics – are fitted on the training partition only. (iii) Class balance is recorded per partition and never silently rebalanced. (iv) A pipeline that fails writes a null-scored record carrying its exception text rather than being dropped.

2.5 A balance-preserving cluster-disjoint splitter

Assigning clusters largest-first drove the test positive fraction to 0.995 on a corpus balanced 1:1 (Section 4.4). We therefore introduce a splitter that keeps clusters **atomic** – never violated, and asserted on real clusters – but changes the *order* of selection: it tracks the remaining per-class quota and repeatedly takes the cluster that best serves whichever quota is further from being met, with seed-dependent jitter so partitions still vary across seeds. Where balance is impossible (all positives in one cluster) it reports the achieved balance rather than promising one.

Because this splitter is post hoc and its effect on our headline statistics is material, it needs to be auditable rather than merely described. Supplementary Figure S1 gives the procedure, Supplementary Table S1 a worked example small enough to check by hand, and Supplementary Table S2 the achieved balance and residual leakage in all 20 cells under both splitters.

Does the result depend on the seed jitter? No. Across the five seeds, the achieved test positive fraction has standard deviation **exactly** 0.000 in all fifteen clustered configurations (Supplementary Table S2); the jitter changes *which* equally-scoring clusters are chosen, not the balance achieved. The unclustered `random` threshold is the only column where the positive fraction varies across seeds at all (0.4985 ± 0.0052), and there it varies because the split itself is random rather than because of the jitter. Residual leakage does vary slightly with seed (standard deviation ≤ 0.012 across all cells), which is expected: the clusters chosen differ, so the above-threshold pairs that straddle the partition differ too.

Stress-testing the splitter against alternatives, and a result against our own method. That the correction raises leakage in some cells invites the obvious question: is any ordering good at all three of test size, class balance and cross-partition similarity? We ran four splitters over identical cached cluster assignments across the fifteen clustered cells, changing only the order in which whole clusters are taken: size-descending (largest-first), quota-driven (ours), **uniformly random**, and a **leakage-aware** greedy order that prefers clusters bridging least to the rest of the corpus.

Table 2: Four cluster-disjoint splitters over the same clusterings, 15 clustered cells. Target test fraction 0.200; lower is better for balance error and residual leakage. **No splitter wins on all three**, which is the trade-off we claimed – but the ordering of the losers is not the one we expected, and it is unfavourable to our own splitter.

	test fraction		balance error		residual leakage	
Cluster order	mean	max	mean	max	mean	max
largest-first (pre-registered)	0.2001	0.2009	0.3045	0.4918	0.1074	0.3928
quota-driven (ours)	0.2001	0.2001	**0.0000**	**0.0000**	0.1237	0.3940
uniformly random	0.2000	0.2003	0.0094	0.0279	0.0548	0.1913
leakage-aware greedy	0.2001	0.2006	0.0425	0.1026	**0.0257**	**0.0530**

Our splitter is the worst of the four on residual leakage, marginally worse even than the largest-first order it was built to replace (0.124 against 0.107 mean), while being the only one to achieve exact balance. The mechanism is now clear and it is a defect of the design rather than of the implementation: both largest-first and quota-driven prefer *large* clusters –

the quota-driven order takes the largest cluster that still fits the remaining quota – and large clusters are precisely the homologous families whose members retain above-threshold similarity to sequences left in training. Ordering by size, for any reason, concentrates leakage.

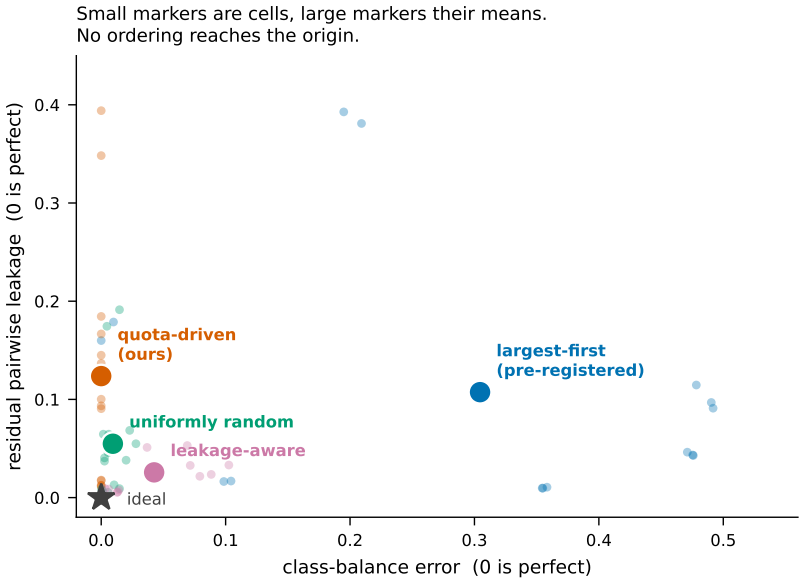


Figure 1: Four cluster-disjoint orderings over identical clusterings. The ideal splitter would sit at the origin and none does, which is the trade-off the text claims. The informative part is where each lands: ordering by size costs class balance (largest-first, far right), our quota-driven order buys exact balance at the highest leakage of the four, and simply shuffling cluster order lands close to the origin on both.

Uniformly random cluster order is the better default (Figure 1). It reduces mean leakage by more than half relative to our splitter (0.055 against 0.124) and its worst case by a factor of two (0.191 against 0.394), at a balance error of 0.009 – not the exact 0.000 the quota-driven order achieves, but far below the 0.305 of largest-first and well inside what any downstream metric would notice. Leakage-aware ordering halves leakage again (0.026 mean, 0.053 worst) at a balance error of 0.043.

We therefore **withdraw the implicit recommendation** that a balance-preserving splitter is the fix, and replace it with the weaker and better-supported claim: the problem is *size-ordered* cluster assignment, and simply shuffling cluster order removes most of both artefacts at once. Our quota-driven splitter remains the right instrument for the sensitivity analysis in this paper, because exact balance is what isolates the class-balance artefact from everything else, and it is used only for that. It is not the splitter we would now advise a benchmark to adopt. No ordering achieved all three targets, so a benchmark must still report which it optimised and what it paid.

2.6 The analysis plan, and every deviation from it

Because this paper argues that evaluation practice is unreliable, it would be self-refuting to analyse it flexibly. We therefore wrote the hypotheses, the decision thresholds and the power analysis before acquiring data, revised them once after the design simulation and before any benchmark was obtained, and report every departure below. We state the status of that document precisely, because overclaiming here would undermine the rest of the paper.

What the plan is, and is not. It is a dated analysis plan, written and revised before execution, released in full with the harness. It was **not** deposited with a third-party registry and carries **no external timestamp**; the only evidence of its ordering is file modification time (plan last revised 2026-10-05; first Arm B results 2026-10-06; Arm A 2026-10-08) and the released history of the harness. We therefore describe outcomes as *pre-specified* rather than appealing

to the stronger guarantee that a registered pre-registration provides, and a reader is entitled to discount them accordingly. We state the thresholds anyway, because a declared threshold that is kept when it produces an unwelcome answer is still worth more than no threshold: ours produced a null, and we report the null.

Pre-specified decision rules. H1: mean pairwise τ_b across evaluation levels is < 0.8, confirmatory if the upper 95% bound is below 0.8. Co-primary: normalised agreement < 0.85 supports H1. Binding power statement: the design has 80% power only for a true mean $\tau_b \leq 0.646$, so an observation in $[0.65, 0.80]$ is reported as inconclusive and not as a near-miss. Top-1 flip rate is descriptive and carries no confirmatory weight. Secondary: H3, the negative-set variance component exceeds the clustering-threshold component; H4, the seed component is under 25% of the largest protocol component; H5, representation ranks are less stable than learner ranks.

Deviations, in full.

1. **The unit of comparison in H1 changed, and this is the consequential deviation.** The plan expressed H1 over *five independently published AMP benchmarks*. That comparison was not executed: the published resources do not share a task definition, a label convention or a redistributable positive set, and the pre-specified gate requiring us to reproduce each benchmark's own reported baseline to within ± 0.03 could not be applied to resources whose headline numbers are not computed on a common target. We replaced it with the two crossed designs described above, in which the compared levels are five negative-set rules (Arm B) and ten biological tasks (Arm A). The estimator, the thresholds and the power statement were carried over unchanged, but **the hypothesis is tested over a different population of evaluation levels than the one originally named**, and the cross-benchmark question in its original form remains open. We flag this as the main reason to treat H1's status as pre-specified rather than pre-registered.
2. **Zoo size 12 → 20 pipelines** (and 16/24 as executed), after the design simulation improved minimum detectable disagreement from $\tau_b = 0.575$ to 0.646. Made before any benchmark was acquired.
3. **Normalised agreement added as co-primary**, with the 0.85 threshold, once the simulation showed τ_b cannot reach 1.0 between seeds of one protocol.
4. **The RBF SVM became an approximation** (Nyström, 512 components) on a measured 9-minute-per-cell exact fit. Validated after the fact in Section 2.10; the plan had listed this as available on request, and we have since done it.
5. **Thread caps and `tree_method="hist"`** for XGBoost, on a measured 97× scheduling artefact. Scheduling only; no model changed.
6. **End-to-end CNN/BiLSTM and ProtT5-XL excluded** as CPU-infeasible.
7. **The balance-preserving splitter and the entire second grid are post hoc**, built after the pre-specified analysis returned a null. Reported as such throughout, and never as confirmatory (Section 4.4).
8. **Residual-leakage measurement was added during execution**, replacing an assertion that failed on real data. It is a measurement we did not anticipate needing, not a changed outcome.

Deviations 2–3 precede data acquisition; 4–6 were forced by timing measurements taken on a single cell with no metric compared across pipelines; 1, 7 and 8 occurred during or after execution and are the ones a reader should weigh.

2.7 The four headline statistics, defined once

Four rank-agreement statistics appear in this paper and they compare different things. To remove any ambiguity about which number refers to what, Table 3 defines each one, names the levels it compares, states how its ceiling is computed, and marks the single **primary** result. Every later mention uses the labels in column 1, and no other agreement numbers are quoted anywhere in the paper.

Table 3: The four rank-agreement statistics. All use mean pairwise Kendall's τ_b on MCC, over pipelines ranked within a cell with ties averaged. The *ceiling* is always the mean Kendall's τ_b between *seed pairs* holding every other factor fixed, so it measures run-to-run reproducibility of the ranking under an unchanged protocol; *normalised agreement* is the ratio of the statistic to its own ceiling. **B1 is the primary result**; B2 is a post hoc sensitivity analysis (Section 4.4); B3 and A1 are secondary. The pre-specified threshold is normalised agreement < 0.85.

Label	Levels compared	n	Splitter	τ_b	ceiling	norm.	Status
B1	negative-set rules	5	largest-first (pre-reg.)	0.727	0.825	**0.881**	**primary**
B2	negative-set rules	5	balance-preserving	0.693	0.867	0.800	post hoc
B3	clustering thresholds	4	largest-first (pre-reg.)	0.817	0.825	0.990	secondary
A1	biological tasks	10	native (as released)	0.479	0.838	0.572	secondary

Ceilings differ between B1/B3 and B2 because the ceiling is recomputed under whichever splitter generated the partitions; it is a property of the protocol, not a constant. B1 and B3 share a ceiling because they share the grid and differ only in which factor is varied.

Reading the primary result. B1 is the pre-registered test and it is a **null**: normalised agreement 0.881 exceeds the 0.85 threshold, so H1 is not supported. B2 is the same comparison after a class-balance defect was corrected (Section 4.4); it falls below the threshold, but it was built after seeing the null and is reported as a sensitivity analysis only. Any statement in this paper that rankings disagree across protocols rests on B1, and B1 does not support it.

2.8 Statistical analysis

Pipelines are ranked within each cell with ties averaged, so we use Kendall's τ_b, which carries the tie correction τ_a lacks. For each pair of levels we report the full matrix, its mean, and the **top-1 flip rate**: the fraction of level pairs naming a different best pipeline. Since leaderboards are read from the top, the latter is the quantity a practitioner experiences.

The agreement ceiling. τ_b does not reach 1.0 even between two seeds of the same protocol, because run-to-run noise reorders near-tied pipelines. We measure that ceiling directly – τ_b between seed pairs within a cell – and report *normalised agreement*, the ratio of between-level τ_b to that ceiling. Reporting raw τ_b alone invites comparison against 1.0, which overstates disagreement. The same reasoning is applied to the top-1 flip rate in Section 5.2, where the ceiling turns out to matter more.

Variance decomposition on two responses. The Arm B grid is a complete balanced factorial with one observation per cell, so method-of-moments variance components are exact in closed form: the expected-mean-square equations solve top-down from the highest-order interaction, which absorbs the residual. We report those as primary and sum-of-squares shares (η^2) alongside them, since the two weight a factor's level count differently and agreement between them is evidence that neither choice drives the conclusion.

We decompose both the metric score and the pipeline *rank* within its cell, because a factor can shift every score without changing any ranking, which is harmless to a leaderboard, or

reorder the top while barely moving scores, which is not. One caveat is essential and we state it where the numbers appear (Section 3.4): because ranks are assigned *within* a cell, every factor that helps define the cell has zero between-group variance on the rank response as an algebraic identity, for any data whatsoever. The rank decomposition is therefore informative about the pipeline main effect and the pipeline-by-factor interactions, and says nothing at all about protocol main effects. Our claims about ordering rest on the rank-agreement and flip-rate analyses instead.

Uncertainty, and the population each interval describes. This deserves to be stated explicitly, because our conclusions and our default intervals generalise over different things.

Resampling **pipelines** answers: *if we had drawn a different set of model pipelines, how much would this agreement statistic move?* It holds the five rules or ten tasks **fixed**, so it is an interval for these particular levels. That is the right population for a claim about the zoo, and the wrong one for a claim about protocols or tasks in general. All intervals labelled "resampling pipelines" are of this kind, and they are our defaults because the pipeline is the unit the statistic is computed over.

Resampling **levels** answers the question our headline claims actually pose: *would another set of negative-set rules, or another set of tasks, give the same answer?* We estimate it with a delete-one-level jackknife, the natural estimator here for two reasons. First, the level is the sampling unit, so deleting one and recomputing propagates exactly the dependence we care about. Second, a bootstrap that resamples levels *with replacement* is badly biased for this statistic: a duplicated level contributes a pair with $\tau_b = 1$ by construction, which inflates agreement upward. Section 5.5 reports these intervals, and they are much wider than the pipeline intervals – with five rules and ten tasks they have to be. We therefore do not make population-wide claims about protocols or tasks, and we say so in the abstract and the conclusion.

Dependence among level pairs. Mean pairwise τ_b over L levels averages $L(L-1)/2$ pairs that are **not independent**: every level appears in $L-1$ of them, so one unusual level perturbs many pairs at once. Treating pairs as independent units understates uncertainty. The pair-level bootstrap in Table 9 does exactly that and is therefore anticonservative; we report it because it is the interval a reader would otherwise compute, and we give the level jackknife beside it as the honest alternative. The level jackknife is immune to this because it resamples levels, not pairs.

Variance-component intervals use a delete-one-seed jackknife, which preserves the balance the decomposition requires. The two arms are compared by a paired bootstrap over the shared pipelines *and*, in Section 5.5, by a two-sample level jackknife; a permutation test swapping rows between arms is unavailable because the arms have ten and five levels respectively.

2.9 Computational notes

Three measured costs shaped the executed configuration, each reported because it is easy to get wrong in the same way we initially did. Sorting sequences by length before batching reduced ESM-2 8M embedding from 959 to 2.1 ms per sequence and 650M from 2008 to 91.6 ms, because unsorted batches pad every member to the longest. Capping worker threads at 8 reduced a gradient-boosting fit on a seven-feature problem from 165 s to 1.7 s, nested parallelism having oversubscribed a 32-thread machine. An exact RBF SVM with probability calibration did not complete one cell in nine minutes, motivating the Nyström approximation. None was chosen after seeing a ranking.

2.10 Validating the kernel approximation

The Nyström substitution is the only change we make to a model rather than to its scheduling, so we validated it directly. We re-ran the physicochemical/SVM pipeline with an exact RBF

kernel and probability calibration across all five negative rules at three seeds (15 matched cells), holding the split, the feature matrix and the hyperparameters fixed, and capping the training set at 2000 sequences so the exact fit was tractable.

Scores track closely. Mean AUROC was 0.799 exact against 0.770 approximate (mean absolute difference 0.044, maximum 0.114); mean MCC was 0.336 against 0.311 (mean absolute difference 0.031, maximum 0.071). Across the 15 cells the two variants correlate at $r = 0.997$ on MCC and $r = 0.969$ on AUROC, with Kendall $\tau_b = 0.810$ and 0.829 respectively. The rule-level ordering, which is the quantity this paper measures, is preserved exactly on AUROC ($\tau_b = 1.000$) and to one adjacent inversion on MCC ($\tau_b = 0.800$). The approximation was 105-fold cheaper (0.6 s against 61.6 s per fit), which is what made the full grid feasible. Per-cell values are in `tables/svm_approximation_check.csv`.

The approximation is therefore slightly pessimistic in absolute score – it costs about 0.03–0.04 of both metrics – but does not reorder the protocol comparison. We treat it as a validated substitution rather than an open caveat.

3 Arm B: varying the evaluation protocol

3.1 The five negative-set rules pose measurably different problems

Positives: 5,187 peptides, median length 18, mean net charge +4.21. Table 4 shows that the rules form an ordered ladder of which shortcuts survive.

Table 4: Corpus characterisation. Matching negative length removes the length artefact exactly but *quadruples* the net-charge gap, because length-matched fragments come from near-neutral protein interiors while antimicrobial peptides are strongly cationic.

Negative-set rule	Neg. median length	Length ratio	Charge gap	Comp. L_1
random UniProt	73	4.06	1.01	1.206
annotation filtered	75	4.17	0.99	1.201
length matched	18	1.00	**4.25**	1.270
composition matched	18	1.00	2.57	0.796
shuffled positives	18	1.00	**0.00**	**0.000**

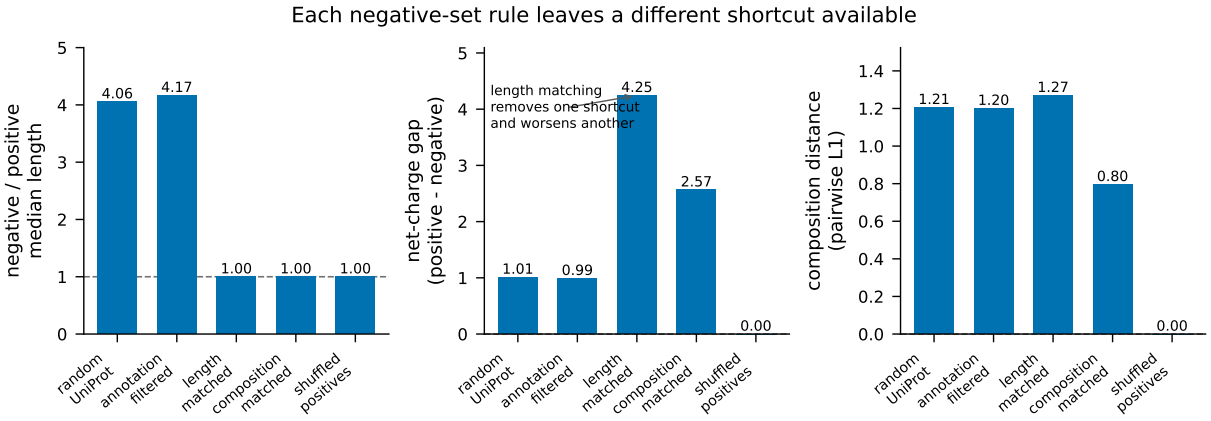


Figure 2: Each negative-set rule leaves a different shortcut available. The middle panel is the counter-intuitive one: the standard remedy for the length artefact makes the charge artefact four times stronger.

3.2 Protocol choice transforms apparent difficulty

Mean MCC over the pipelines spans **0.311 to 0.913** across the twenty protocol cells (Figure 3). The same pipelines on the same positive set produce results from barely-better-than-chance to near-perfect, depending on choices a methods section usually compresses into one sentence.

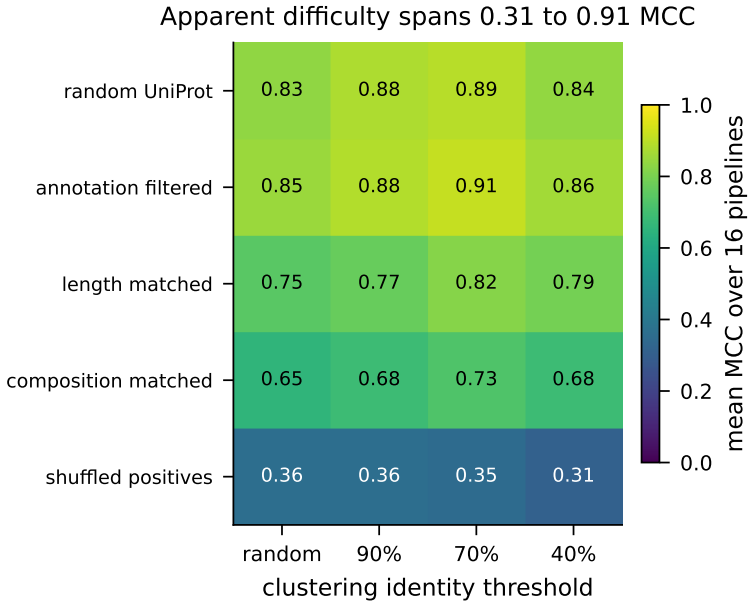


Figure 3: Mean MCC by negative-set rule and clustering threshold.

3.3 The ordering is largely preserved; the winner is not

Two different questions, kept apart. "Do protocols agree?" has two answers that point in opposite directions, and conflating them is how this literature reaches overconfident conclusions in either direction. We therefore report them separately throughout:

- **Broad ordering** – does the whole leaderboard keep its shape? Measured by mean pairwise τ_b against its seed ceiling. Answer: *largely, but not entirely.*
- **Winner identity** – does the same pipeline come first? Measured by the top-1 flip rate against its own seed ceiling (Section 5.2). Answer: *often not.*

On the primary analysis **B1**, mean pairwise τ_b across the five rules is 0.727 against a measured seed ceiling of 0.825: normalised agreement **0.881**, *above* the pre-specified threshold of 0.85, so H1 is **not supported**. Under the post hoc **B2** the same comparison gives $\tau_b = 0.693$ against a ceiling of 0.867, normalised 0.800 (Section 4.4; labels as in Table 3).

The broad ordering is therefore stable but *not* fixed, and we want to be careful not to overstate the stability either. A normalised agreement of 0.881 means roughly one-eighth of the reproducible ordering is lost to protocol choice, and the largest rank swing is 7 of 16 positions (Figure 5) – rankings do change. What does not happen is wholesale scrambling: the weakest pipelines stay at the bottom under every rule.

Winner identity behaves differently. **9 of 10 rule pairs name a different best pipeline**, with four distinct winners. This is not in conflict with the high τ_b: τ_b weights all 120 pairwise orderings equally and most are stable, while the leading places – which are close in score – reorder. Section 5.2 shows how much of that reordering exceeds run-to-run noise, and Section 5.5 how precisely five rules can locate it.

Clustering threshold leaves the *ordering* untouched. Under the pre-registered splitter $\tau_b = 0.817$ against a ceiling of 0.825, a normalised agreement of 0.990; under the balance-preserving

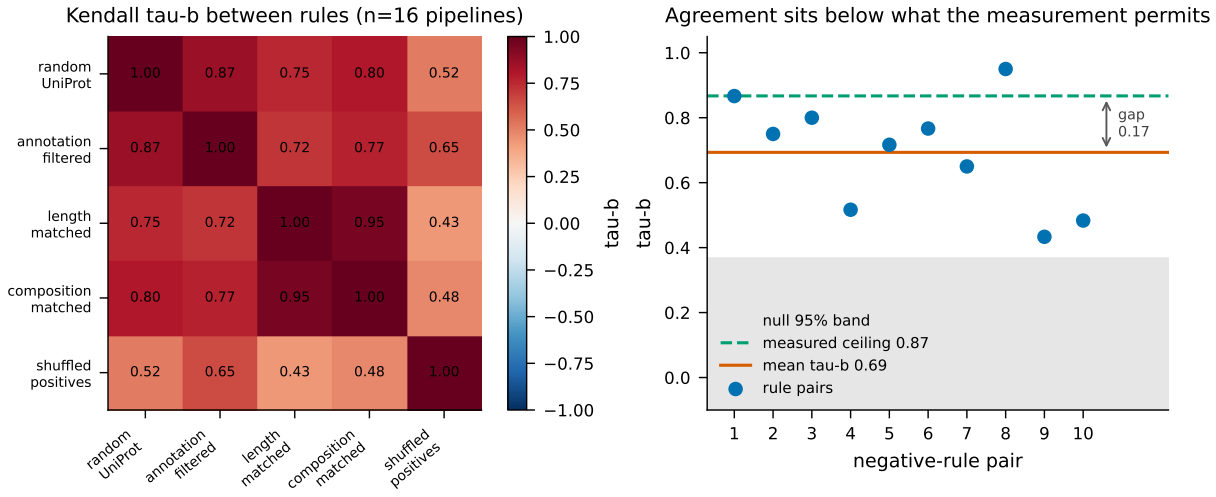


Figure 4: Left: τ_b between every pair of negative-set rules. Right: the ten pair values against the *measured* seed ceiling and against the permutation null band. Agreement sits well above chance and clearly below what the measurement permits – the gap the normalised statistic quantifies. This, not the rank-variance decomposition, is the direct evidence for the ordering claim.

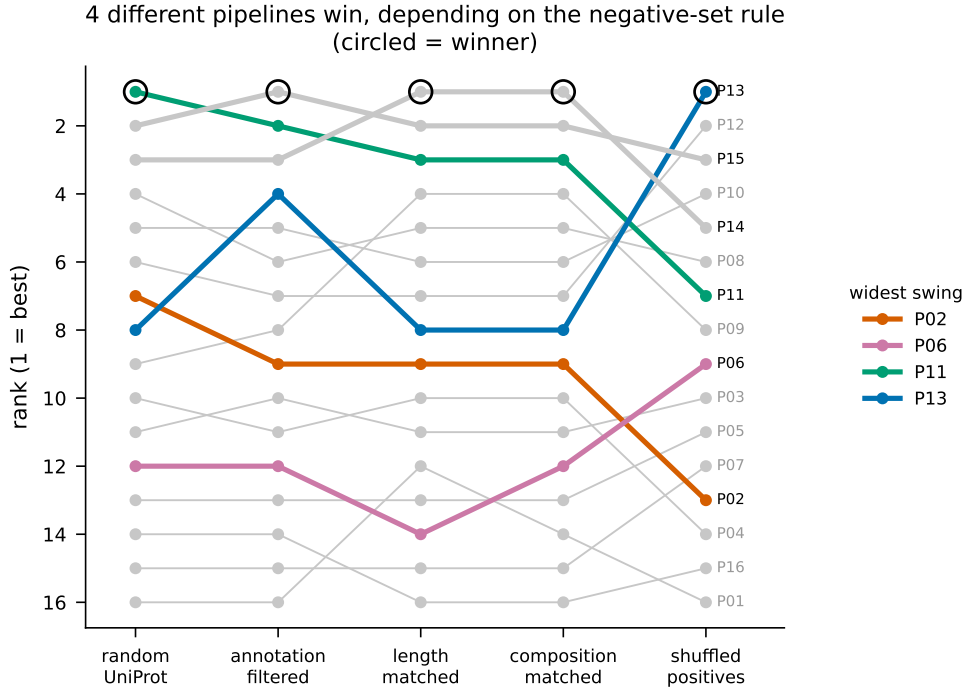


Figure 5: Pipeline rank against negative-set rule. Circled markers are the winner under each rule.

splitter $\tau_b = 0.889$ against 0.867, i.e. normalised 1.03 – at or above the ceiling, meaning the threshold reorders pipelines no more than reseeding the same protocol does.

Its effect on the *winner* depends on how much averaging is done first, and we were initially too quick here. Scoring each pipeline on its mean over all five seeds and all five rules, no threshold pair changes the winner (0 of 6). But a published benchmark reports one run, not a mean over twenty, and at single-run resolution – holding the rule and seed fixed and comparing thresholds – the winner changes in 75.3% of pairs, against a seed-noise ceiling of 54.5% (difference +0.208, $[+0.110, +0.305]$ over pairs; Section 5.2). That is close to the negative rule's 76.0%. Both figures are descriptive of these four thresholds and five rules: the interval resamples pairs drawn from a handful of levels, and resampling the levels instead leaves the threshold flip rate wholly undetermined, $[0.000, 1.000]$ (Section 5.5).

We therefore make the narrower claim the data support: **within this corpus the threshold is benign for the ordering but not for the winner**, and the negative-set rule is the factor that damages both. Read as an observation about current practice rather than as an inference, the field has standardised the factor that is harmless to the ranking and left unstandardised the factor that is not.

3.4 What moves the score is not what moves the ranking

Score: the negative-set rule owns most of the variance

Table 5 decomposes the MCC of the complete balanced factorial (16 pipelines × 5 rules × 4 thresholds × 5 seeds, one observation per cell) two ways. The negative-set rule is the largest single component on either measure: $\hat{\sigma}^2 = 0.0467$, **71.0%** of the total estimated variance, against 14.8% for the pipeline itself. **Which negatives a benchmark invents matters more to the number it reports than which model is run on it.** The clustering threshold accounts for 0.6% and the seed for a component estimated at -1×10^{-6}, i.e. indistinguishable from zero.

The two columns are different estimators of the same thing and agree closely, which is why we show both. Moment-based variance components put each factor on a common footing regardless of how many levels it has; sums-of-squares shares (η^2) do not, and favour factors with more levels. Agreement to within four percentage points means the conclusion is not an artefact of either choice. We report the moment estimates as primary: the design is a complete balanced factorial, so the expected-mean-square equations solve exactly in closed form, whereas a REML fit of the same crossed model lands on the parameter boundary and does not converge.

Table 5: Variance decomposition of MCC in Arm B. Columns 2–3 are exact method-of-moments variance components; column 4 is the sum-of-squares share of the same model, shown to demonstrate the two estimators agree. Column 5 is the sum-of-squares share on the *rank* response, with delete-one-seed jackknife intervals. Protocol main effects are *structurally* zero on rank – see the text; this is an identity, not a measurement.

	score (MCC)			rank	
Component	$\hat{\sigma}^2$	% $\hat{\sigma}^2$	% SS	% SS	95% CI
negative rule	0.0467	**71.0**	67.9	0†	–
clustering threshold	0.0004	0.6	0.7	0†	–
threshold × rule	0.0003	0.5	0.4	0†	–
seed	$< 10^{-5}$	0.0	0.0	0†	–
pipeline	0.0098	14.8	19.2	83.47	[82.83, 84.11]
pipeline × rule	0.0073	**11.0**	10.0	**8.87**	[8.30, 9.43]
pipeline × threshold	0.0005	0.8	0.8	1.26	[0.97, 1.55]

†Identically zero for any dataset, because ranks are assigned within a cell: see Section 3.4.

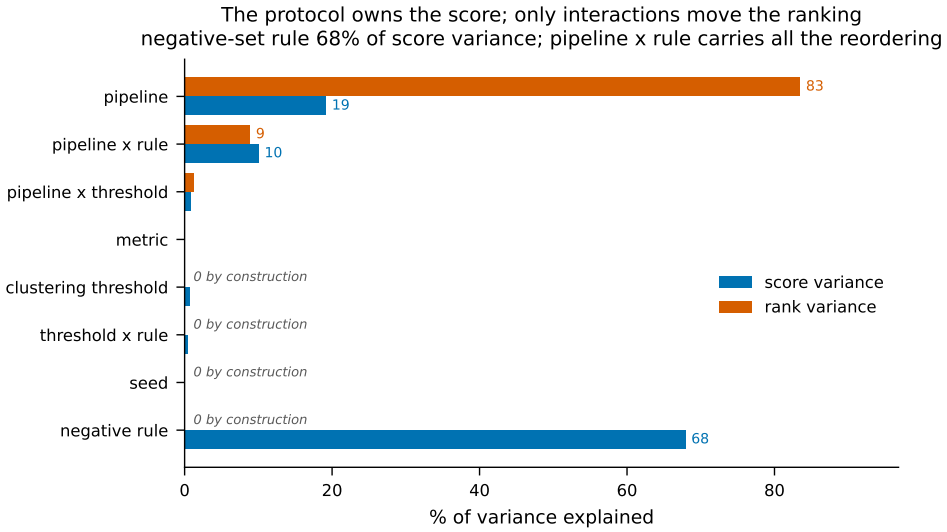


Figure 6: Score variance against rank variance by component.

Rank: why the protocol main effects are zero by construction

Repeating the decomposition on the within-cell rank returns *exactly* zero for every protocol main effect. We initially read this as a striking empirical result. It is not: it is an algebraic identity, and we state it as one because the arithmetic is worth having in the open.

Ranks are assigned to the K pipelines separately within each (metric, threshold, rule, seed) cell. Every cell therefore contains each rank exactly once, so the mean rank in every cell is $(K+1)/2$ regardless of the scores. Any factor that helps *define* the cell partitions the data into groups of whole cells, so each of its group means is also $(K+1)/2$, and its between-group sum of squares is identically zero. We verified this numerically – the mean rank is 8.500 for all five rules, all four thresholds and all five seeds, with $K = 16$ – but no verification was needed. The jackknife interval $[0.00, 0.00]$ and standard error of 0.00 we previously reported are the signature of a constant, not evidence of a precisely measured null.

This is not a defect of the rank response; it is the reason rank-based analysis is the right instrument. A factor that shifts every score in a cell by the same amount cannot touch the ordering, and the decomposition records that by construction. But it means the rank column cannot be used to argue that protocol choice *happens* not to reorder pipelines, and we no longer do so. On the rank response the informative terms are the pipeline main effect and the pipeline-by-factor interactions, which are free to vary and do:

- pipeline × rule accounts for 8.87% $[8.30, 9.43]$ of rank variance, **seven times** the pipeline × threshold term (1.26% $[0.97, 1.55]$). All reordering attributable to the protocol is carried by these interactions, and the negative-set rule dominates the threshold within them.
- On the score response the same two terms are 11.0% and 0.8% – the only components whose relative standing is similar across the two responses, because an interaction is precisely a term that moves pipelines differentially.

The substantive claim of this section therefore rests on the direct rank-agreement evidence of Sections 3.3 and 5.2, not on the zero entries in Table 5. What the decomposition does establish is the *score* result: a single protocol choice owns 71% of the variance in reported performance, seven times the pipeline's own 14.8%.

3.5 The null baseline tracks the shortcut ladder

A two-feature pipeline (length, net charge) reaches MCC 0.793 on proteome-derived negatives, within 0.11 of a 650M-parameter language model, and falls to **−0.074** – below chance – on shuffled positives, where length, charge and composition are identically zero by construction (Figure 7). The progression was predicted from Table 4 before the grid was run.

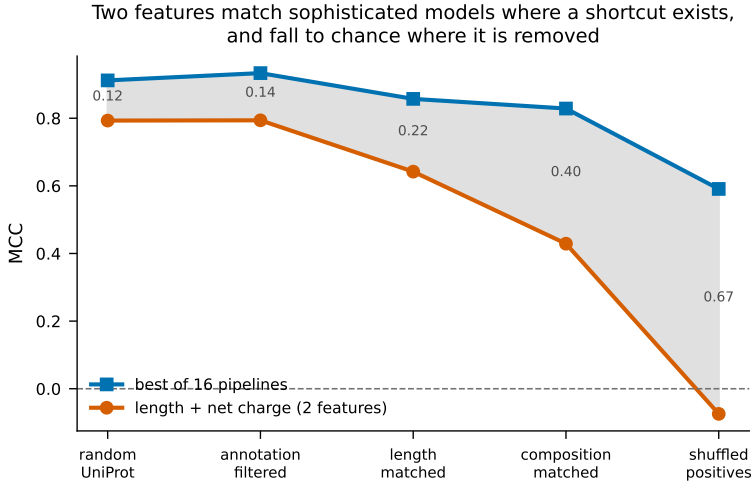


Figure 7: The two-feature baseline against the best of the zoo, by negative-set rule.

A benchmark whose negatives are drawn as whole proteome sequences is substantially solvable without any model of antimicrobial activity.

3.6 Class provenance is asymmetric under two of the five rules

Two of our five negative-set rules draw their negatives *from* reviewed UniProt, while the positives come from DBAASP. The two classes therefore differ in provenance, not only in activity, and that raises a question for any model pretrained on proteomes: is one class made of ordinary proteome entries and the other not?

Table 6 measures it. Under those two rules the asymmetry is close to total: **100% of negatives and** 7.9% **of positives** are verbatim entries in the reviewed UniProt release we retrieved, a 92-point gap aligned with the class label. Under the three rules that synthesise negatives from the positives – length matched, composition matched, shuffled – negative overlap falls to 0.06% or below and the gap nearly vanishes.

This is an overlap audit against current UniProt, not a measurement of ESM-2's training data. We are explicit about that because the tempting stronger claim is not supported, and an earlier draft of this section made it. ESM-2 was pretrained on UniRef derived from UniProt release **2021_04** [15]; our audit uses reviewed UniProt retrieved **2026-10-05**. Overlap with the latter neither implies nor excludes presence in the former, and it fails in *both* directions:

- a sequence in the 2026 release may have been deposited after 2021_04 and so was never available to the model, so overlap is not a lower bound on exposure;
- UniRef is built from the whole of UniProt, whereas we checked only the *reviewed* subset – about 4.9×10^5 sequences against the order of 10^8 in UniProt overall – so a sequence absent from our audit may still have been in training, and non-overlap is not an upper bound either.

Establishing exposure would mean reconstructing the 2021_04 UniRef50/90 release and checking membership there, cluster-representative status included. We have not done that.

What the audit does establish is a fact about the benchmark rather than about the model: **under the two proteome-derived rules one class consists entirely of ordinary reviewed-proteome entries and the other almost entirely does not.** Any encoder pretrained on proteomes is then asked to separate two classes that differ in provenance as well as in antimicrobial activity – a route to a high score that is not a model of the property, and one available independently of the length and charge shortcuts.

Table 6: Exact sequence membership in reviewed UniProt as retrieved 2026-10-05, by negative-set rule and class. This is an overlap audit against the current release, not a reconstruction of any model's training set. The positive set is identical across rules by construction, so its 7.9% is a constant; what varies is the negatives.

	negatives in UniProt		positives in UniProt	
Negative-set rule	n	%	n	%
random UniProt	5,187	**100.00**	410	7.9
annotation filtered	5,187	**100.00**	410	7.9
composition matched	3	0.06	410	7.9
length matched	1	0.02	410	7.9
shuffled positives	0	0.00	410	7.9

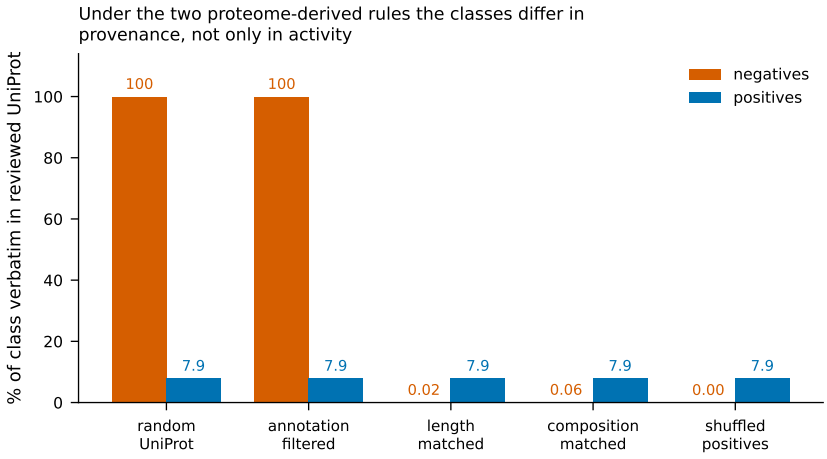


Figure 8: Percentage of each class that is a verbatim entry in reviewed UniProt as retrieved 2026-10-05. The positive set is fixed across rules, so its 7.9% is a constant and only the negatives move. Under the two proteome-derived rules the classes differ almost completely in provenance; under the three synthesised rules they do not. This is an overlap audit against the current release, not a measurement of any model's training set.

A provenance shortcut, independent of length and charge. The two rules whose negatives are wholly proteome-derived are the same two on which the two-feature baseline reaches MCC 0.79 and mean MCC reaches 0.91. A benchmark built that way offers two unrelated routes to a high score – the compositional shortcut of the ladder above, and discriminating proteome entries from synthetic peptides on whatever regularities distinguish them – and neither is a model of antimicrobial activity. We cannot separate the two routes with this design, and we make no claim about what the ESM-2 pipelines actually use; the point is that the benchmark cannot distinguish them either.

The same audit on Arm A. Of 9,655 distinct test sequences across the ten tasks, 12.1% are verbatim reviewed-UniProt entries, and the rate varies enormously by task: **toxicity** 55.8%, antifungal 26.3%, antimicrobial 11.9%, neuropeptide 11.7%, down to DPP-IV 1.2% and ACE-inhibitory 1.9% (`tables/esm_exposure_audit.csv`). Toxicity is also the task where cluster-

disjoint splitting inverts the sign of net charge (Section 4.4c), so it is at once the most proteome-like and the most shifted of the ten. Because provenance varies across tasks by a factor of 45 it is a plausible contributor to the poor rank transfer of Arm A, and it is not something any leaderboard currently reports. The same caveat applies as above: this describes what the datasets are made of, not what any model was trained on.

3.7 Would an equal tuning budget change the answer?

Fixing hyperparameters across every cell is what keeps protocol preference from being confounded with search effort, but it leaves a fair objection standing: a fixed setting may suit a random forest better than an MLP, so part of the ranking could reflect whose defaults were luckier rather than whose method is better.

We tested it by giving every pipeline the **same budget** – six candidate configurations for each of the five learners – and selecting among them on a validation split taken from inside the training partition. Two properties are enforced: the inner split is **cluster-disjoint**, using the clustering that produced the outer split, so hyperparameters are not chosen on sequences homologous to those they are validated against; and the **test partition is untouched** during selection, scored once after refitting the chosen configuration on the full training partition. Five rules at threshold 0.7, seed 0, the 16 Arm B pipelines: 560 fits.

Table 7: Equal tuning budget against the fixed hyperparameters used throughout. Agreement across rules is the quantity the paper's conclusions rest on, and tuning does not move it. The within-rule comparison is given against the seed ceiling for the same design, which is the right scale for judging it.

	fixed	equal budget
mean test MCC over 80 pipeline-rule cells	0.487	0.491
mean pairwise τ_b *across rules*	0.563	0.587
within-rule, fixed ranking against tuned ranking		
mean τ_b	0.843	(min 0.783)
seed ceiling, same design		0.867
rules whose winner changes		3 of 5

The conclusions do not change. Agreement across the five rules is 0.563 with fixed hyperparameters and 0.587 with an equal budget: tuning does not rescue cross-protocol agreement, which is the claim that would have been in danger. The mean gain from tuning is +0.005 MCC, with a maximum of +0.056 – the fixed settings were close to the best of six for most pipelines.

There is a mild family bias, in the direction the objection anticipated. Mean gain by learner is +0.017 for the approximate SVM and +0.010 for gradient boosting, against +0.004, +0.003 and +0.002 for logistic regression, random forest and the MLP. The fixed defaults suited the kernel machine and the booster least. The effect is real and we report it, but it is about 0.02 MCC, against the 0.6 MCC that separates the easiest protocol cell from the hardest: a thirtieth of the effect the paper is about.

Tuning does move the winner, and the reason is the paper's own thesis. Three of five rules crown a different pipeline under tuning, and the fixed-versus-tuned ranking agreement, $\tau_b = 0.843$, is slightly below the seed ceiling of 0.867 for the same design – so tuning perturbs the ranking about as much as reseeding does, marginally more. Inspecting the three changes shows what they are: under annotation-filtered negatives the winner moves from P14 at MCC 0.960 to P10 at 0.958, under composition-matched from P10 at 0.340 to P13 at 0.342, under length-matched from P08 at 0.430 to P12 at 0.439. These are near-ties decided by the third decimal. They do not show that tuning changes which method is better; they show again that the top of this leaderboard is a near-tie that any perturbation reorders (Section 5.2).

3.8 Metric choice reorders models about as much as the negative rule

Mean pairwise τ_b between the six metrics, protocol averaged out, is 0.780 – comparable to the 0.693 between negative rules. Accuracy against AUPRC is 0.617. Entered as an explicit factor, metric accounts for 31.1% of score variance, exceeding the negative-set rule.

3.9 Is it the rule, or the particular negatives it happened to draw?

The main grid embeds **one** negative realisation per rule, because embedding five draws at four ESM-2 scales was not affordable on CPU. The reported seed component therefore varies the partition and the model initialisation but not *which* negatives were sampled, and a rule effect estimated from a single draw could in principle be that draw's idiosyncrasy rather than the rule's.

We close this on the twelve embedding-free pipelines, where no ESM cache is needed: five independent negative pools per rule, 25 cells at the 0.7 threshold, with the seed held fixed so the draw is the only thing moving. The seed ceiling and the across-rule value are recomputed on the *same* twelve pipelines at the same threshold, so all three numbers in Table 8 are comparable; they are not comparable to the 16-pipeline statistics elsewhere in the paper, and we do not mix them.

Table 8: Three sources of ranking movement, measured on one matched design: 12 embedding-free pipelines, threshold 0.7. *Normalised* divides by the seed ceiling, which is itself computed on these twelve pipelines. The negative-set rule disturbs the ranking roughly twice as much as redrawing the negatives within a rule, which is the reassurance the main grid needed; but the draw is not negligible, and for the two matched rules it approaches the rule effect.

What varies	mean τ_b	normalised
seed only: partition and initialisation (the ceiling)	0.857	1.000
negative draw, within a rule	0.728	0.849
negative rule	0.357	**0.417**
per-rule draw agreement		
shuffled positives	0.867	1.011
random UniProt	0.818	0.955
annotation filtered	0.806	0.940
length matched	0.590	**0.688**
composition matched	0.559	**0.653**

The main conclusion survives: the rule dominates the draw, by roughly a factor of two on the normalised scale (0.417 against 0.849; Figure 9a). Fixing the realisation did not manufacture the disagreement we attribute to the negative-set rule.

But the draw is a real source of variation, and it is concentrated in exactly the rules the field would consider most careful. For shuffled positives and the two proteome-derived rules, redrawing the negatives leaves the ranking essentially at the ceiling (0.94–1.01 normalised): those constructions are stable because they are either deterministic given the positives or sampled from a pool so large that any two draws look alike. For **length matched** and **composition matched** – the rules designed to remove the length and composition shortcuts, and therefore the ones a careful benchmark is most likely to adopt – normalised draw agreement falls to 0.688 and 0.653. Each matched negative is chosen per-positive from a candidate pool, so two draws under the same rule yield materially different problems.

Two consequences we would rather state than have inferred. First, for those two rules the single-draw rule effect reported in the main grid is partly draw-specific, and the honest reading of a matched-negative benchmark is that *the rule does not determine the benchmark* – the realisation has to be published alongside it. Second, our seed component understates total run-

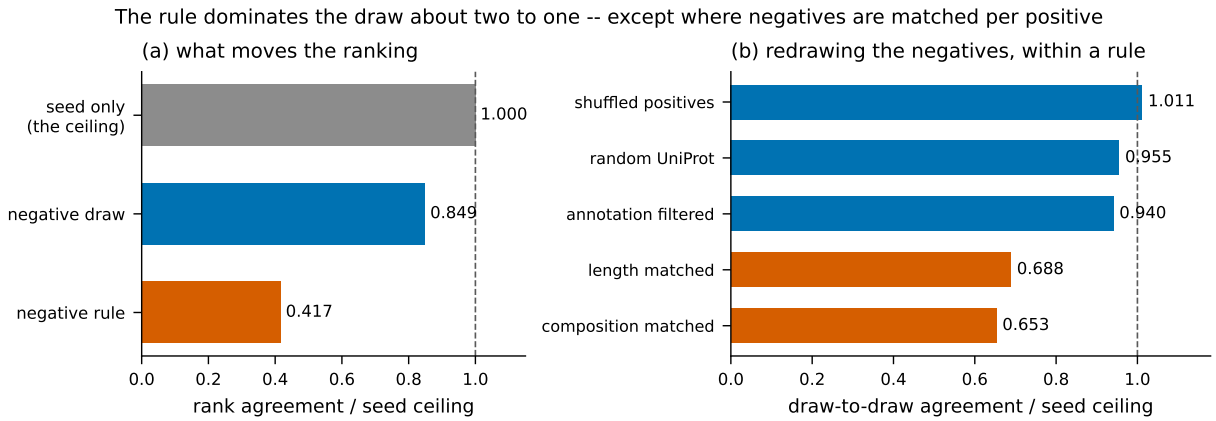


Figure 9: (a) The negative-set rule disturbs the ranking about twice as much as redrawing the negatives within a rule, both read against the seed ceiling. (b) That average hides the important part: for the three rules whose negatives are either deterministic or drawn from a very large pool, redrawing changes almost nothing, while for the two *matched* rules – the ones designed to remove the length and composition shortcuts – the draw alone moves the ranking to 0.65–0.69 of the ceiling.

to-run variability for those rules, because the draw sits outside it; the decomposition in Table 5 should be read as conditional on the realisation rather than marginal over it. A benchmark reporting a matched-negative result from one draw is reporting something less reproducible than its seed variance suggests.

4 Arm A: varying the biological task

4.1 Rankings transfer poorly across tasks

Holding the protocol fixed and varying the task gives mean pairwise $\tau_b = 0.479$ $[0.351, 0.564]$ against a measured ceiling of 0.838, a normalised agreement of **0.572**. The null standard deviation of the mean statistic for this design is 0.022, so this is far from chance and equally far from transfer.

42 of 45 task pairs name a different best pipeline (flip rate 0.933 $[0.800, 0.956]$), with **seven distinct winners** across ten tasks and a largest rank swing of **20 of 24 positions**. A method selected on antimicrobial data carries little information about which method will win on toxicity.

4.2 Difficulty varies as much across tasks as across protocols

Mean MCC over 24 pipelines ranges from 0.608 (tumour T-cell antigen) to 0.254 (anticancer), with antimicrobial at 0.255 – far below the 0.91 the same pipelines reach on proteome-derived negatives in Arm B. The difference is not the methods; it is that these benchmarks use curated negatives rather than arbitrary proteome sequences. This is the clearest demonstration of the Arm B finding: **the negative-set rule, not the method, sets the achievable number.**

4.3 The null baseline behaves oppositely here, and that is the point

Across all ten tasks the two-feature baseline ranks 20th to 24th of 24, with margins to the best pipeline from 0.17 to 0.68. On proteome-derived negatives in Arm B the same two features reached 0.79. **The shortcut was never a property of peptide prediction; it was a property of how the negatives were drawn.**

4.4 Three further artefacts of homology-aware splitting

(a) Cluster-disjoint is not pairwise-disjoint

Greedy clustering does not form equivalence classes, so two sequences can occupy different clusters and still exceed the identity threshold. Measured in-grid, the fraction of test sequences with an above-threshold hit in training rises from 1.41% at a 90% threshold to **21.63**% at 40%, reaching 39% for whole-protein negatives across 45 independent splits. **The stricter the nominal threshold, the more leakage survives**, because a loose criterion admits far more qualifying pairs and short peptides clear 40% identity by chance. The effect is unchanged under the balance-preserving splitter (22.99% at 40%), confirming it is a property of cluster-disjoint splitting itself.

(b) It destroys class balance

On a corpus balanced 1:1 by construction, test-set positive fraction reaches **99.5**% at the 40% threshold. The mechanism is rule-dependent: antimicrobial peptides are homologous to one another and form large clusters, so largest-first assignment sends them to the test partition, while negatives derived *from* the positives remain singletons. Whole-protein negatives form their own families and occupy the test partition instead, inverting the imbalance to 0.29.

Our balance-preserving splitter (Section 2.5) restores the test positive fraction to 0.500 in **all fifteen** rule-by-threshold configurations, holds the test fraction at exactly 0.200, and never splits a cluster. Re-running the full grid with it **strengthens every finding**: normalised agreement falls 0.881 → 0.800, winner flips rise 7/10 → 9/10, and the pipeline-by-rule rank-variance share rises 6.9% → 8.9% (Figure 10). The imbalance had been *masking* disagreement, because a 99%-positive test set compresses MCC and makes pipelines bunch.

Status of this analysis. The splitter was built after observing that the pre-registered analysis did not support our hypothesis. Its motivation was a verification failure logically independent of the ranking result, but the ordering of events is what pre-registration guards against. We therefore report the pre-registered analysis as primary and this one as **post hoc**, and do not claim the hypothesis confirmed.

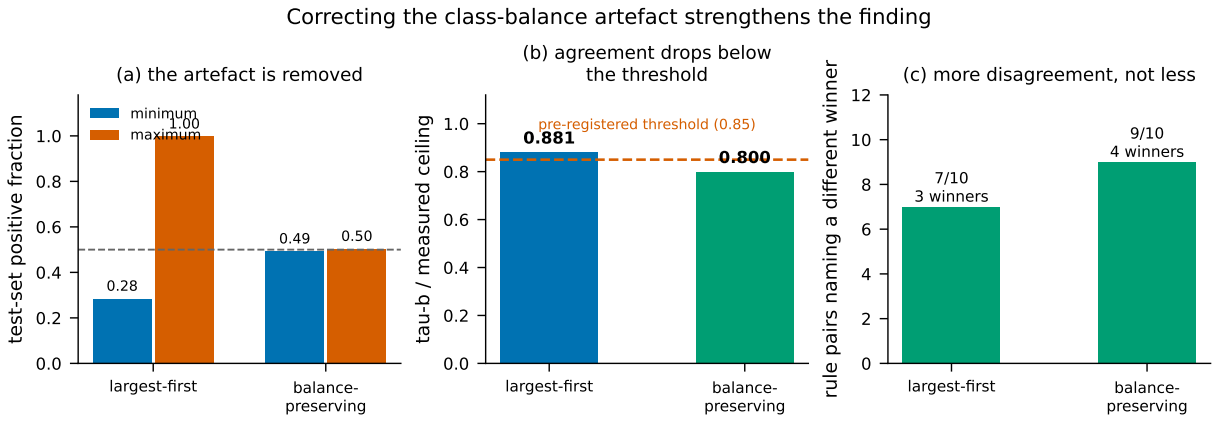


Figure 10: Correcting the class-balance artefact removes it entirely and strengthens, rather than weakens, the ranking finding.

(c) It induces covariate shift, sometimes reversing a feature

Held-out clusters are by construction the sequences least similar to the training set, so the held-out partition can differ systematically in exactly the features a model relies on. Net-charge

separation between classes shifts from +1.80 (train) to **−1.48** (test) in toxicity and from +1.18 to −0.12 in antifungal: **the relationship between charge and the label inverts between partitions**. Positive-class median length shifts by up to 40 residues. The shift is associated with apparent difficulty (Spearman $\rho = -0.624$, $p = 0.054$ for charge-gap loss; $\rho = -0.564$, $p = 0.090$ for composition shift); with ten tasks these are suggestive rather than established, though the sign reversals are directly observable.

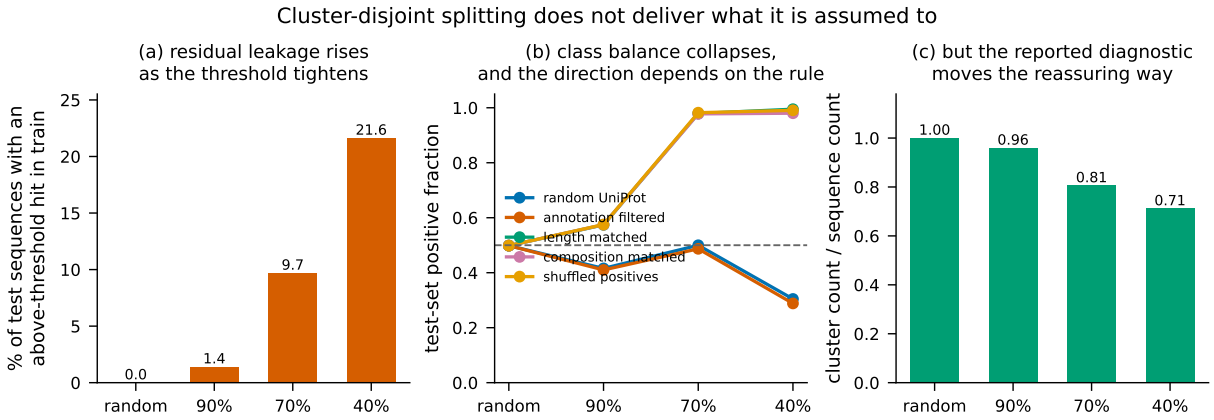


Figure 11: Three artefacts of cluster-disjoint splitting. The diagnostic a benchmark would naturally report (panel c) moves in the opposite direction to both problems.

(d) Published benchmarks contain exact duplicates across their own splits

Of the 16 datasets in the collection, **11 contain sequences present in both partitions**, listed in full in Supplementary Table S3. These are byte-identical strings, which any similarity threshold must treat as 100% identical; we remove the test-side copy before scoring and report the count. Separately, sequences appear *with both labels* inside a single partition, most heavily in antibacterial (295 in train, 31 in test).

Which splits were used for the reported runs. Arm A uses each dataset's *released* `train.csv`/`test.csv` partition unchanged, with the test-side copy of any cross-split duplicate dropped before scoring and no other modification; the five seeds vary model initialisation and data order, not the partition, which is fixed by the publisher. Arm B generates its own partitions, and every Arm B number is reported under one of two named splitters (largest-first or balance-preserving, Table 3), never a mixture.

We note these neutrally. They are observations about released files, and they were only checkable because the authors released them. The point is not that this collection is careless – it is unusually careful, and the duplicates survived anyway.

5 Uncertainty, and a direct test of the two arms

5.1 Headline statistics with intervals

The point estimates, ceilings and normalised values for all four statistics are given once, in Table 3, and are not repeated here. This section adds the uncertainty: Table 10 below carries both kinds of interval side by side, computed in one run of each estimator (3,000 resamples for the pipeline bootstrap, ranks recomputed inside each resample).

Marginal top-1 flip rates, computed on seed-averaged scores, are 0.933 $[0.800, 0.956]$ in Arm A over 45 pairs and 0.900 $[0.400, 0.900]$ in Arm B over only ten; both intervals resample pairs and are therefore anticonservative, and the Arm B one is too wide to support a conclusion on its

own. Sections 5.2 and 5.5 replace both with a matched contrast against a measured ceiling and with level-based intervals.

5.2 The flip rate against its own ceiling

Section 3.3 insists that τ_b be read against a measured ceiling rather than against 1.0. The same argument applies to the flip rate, and applying it is the sharper test: reseeding alone reorders near-tied pipelines, so some winner changes are run noise rather than protocol sensitivity. Quoting a flip rate against an implicit floor of zero makes exactly the mistake the ceiling was introduced to prevent.

We therefore measure the ceiling the same way, and make the contrast matched on seed – across levels with the seed held fixed, against across seeds with the level held fixed. Averaging over seeds before taking the winner, as the marginal flip rates above do, suppresses run noise in the first term but not the second, so the two would not otherwise be comparable.

Table 9: Top-1 flip rate against its measured seed-noise ceiling, matched on seed. *Normalised excess* is (observed − ceiling)/(1 − ceiling): the share of the room above the noise floor that the factor actually uses. Bootstrap over pairs, 20,000 resamples.

Factor varied	arm	flip	ceiling	difference (95% CI)	norm. excess
biological task	A	0.933	0.570	+0.363 [+0.262, +0.467]	**0.845**
negative-set rule	B	0.760	0.545	+0.215 [+0.125, +0.305]	0.473
clustering threshold	B	0.753	0.545	+0.208 [+0.110, +0.305]	0.458
negative-set rule‡	B	0.730	0.585	+0.145 [+0.055, +0.235]	0.349

‡Under the pre-registered splitter; the three rows above use the balance-preserving splitter. All four differences have $p \leq 0.002$.

Three things follow, and the first is a correction to our own emphasis.

The winner is unstable even to reseeding. The ceiling is 0.545 in Arm B and 0.570 in Arm A: *more than half* of all winner changes are reproduced by changing nothing but the random seed. A benchmark that reports a single run and names a best method is, on this evidence, reporting a coin flip about as often as not. This is a stronger practical warning than anything we had framed as a headline, and it is independent of protocol choice entirely.

Within these levels, protocol choice flips the winner more often than noise does – but this is descriptive, not an inference about protocols in general. Every factor exceeds its ceiling in the table above, and the pre-registered splitter still shows an excess (+0.145, [+0.055, +0.235]), so the pattern does not depend on the post hoc correction of Section 4.4. We nevertheless decline to call the Arm B version significant, for a reason that is structural rather than cautious: **the intervals in Table 9 resample pairs, and the ten rule pairs are built from only five rules**, each appearing in four pairs. They are not independent evidence, so a bootstrap over pairs understates uncertainty. Resampling the rules themselves gives a level-based interval on B1's flip rate of [0.060, 1.000] (Section 5.5) – five levels cannot locate that rate at all.

So we make two claims of different strength. *For these five negative-set rules*, the winner changes in most pairs and more often than reseeding alone changes it: that is a measurement, and it is the experience of a practitioner switching between two published negative-set definitions. *For negative-set rules in general*, we make no claim; the data do not support one and we have not smuggled it in through a p-value computed over pairs.

Task variation again dominates, and there the claim does generalise. Changing the biological question consumes 84.5% of the available room above the noise floor, against 47.3% for the negative-set rule. With ten tasks rather than five rules the level-based interval on the task flip rate is [0.853, 1.000], so unlike Arm B this one survives resampling of the levels – still within the one curated collection those ten tasks come from.

5.3 The two arms differ, and task variation is the larger effect

Both arms rank the same pipelines, so the comparison is paired at the pipeline level: we resample pipelines once per iteration and recompute *both* τ_b values on that same resample. A permutation test swapping rows between arms is unavailable, since the arms have ten and five levels respectively and the rows are not exchangeable; we attempted it, it failed on the dimension mismatch, and we replaced it rather than forcing a test the data do not support.

Arm B splitter	pre-registered	balance-preserving
τ_b, varying protocol	0.727	0.693
τ_b, varying task (shared pipelines)	0.508	0.508
difference	**+0.218**	**+0.185**
95% CI	[+0.117, +0.317]	[+0.110, +0.269]
two-sided p	0.001	0.00025

Across these five rules and ten tasks, changing the biological task disturbs the ranking more than changing the evaluation protocol. Neither arm alone could have established this.

Two qualifications belong with that sentence rather than in a limitations list. First, the intervals above resample pipelines and so hold the levels fixed; generalised over levels the difference is $+0.218$ $[+0.029, +0.408]$, $p = 0.024$ under the primary analysis **B1** and $+0.185$ $[-0.061, +0.431]$, $p = 0.141$ under the post hoc **B2** (Section 5.5). The claim rests on B1. Second, Arm A varies the biological question and not the curator, so this contrasts *tasks within a protocol* (Section 6.1).

5.4 Cluster-aware intervals on the scores

The intervals above are all on rank statistics. The scores have their own uncertainty, and the usual sequence-level bootstrap is optimistic for peptides because test sequences are related to one another and are therefore not independent draws. We recomputed key score intervals resampling whole similarity clusters instead, which requires refitting with the predictions retained – the stored run records hold aggregate metrics only – so we scoped it to the five negative rules at threshold 0.7 (16 pipelines) and the five smallest Arm A tasks (24 pipelines), the latter being where a sequence-level interval is most optimistic. For Arm B the resampling unit is the same clustering that defined the split; for Arm A, which ships a partition with no published clustering, the test set is clustered at 40% identity, the coarse end, chosen because coarser clusters give wider intervals.

In Arm B the sequence-level interval is too narrow, by $1.63\times$ **on average and** $2.64\times$ **at worst** (Figure 12), and the correction is largest for annotation-filtered and random UniProt. The mechanism is visible in the table: whole proteome sequences form families, so those test partitions hold 6.5–6.7 sequences per cluster, while negatives synthesised from the positives are near-singletons at 3.2–3.3. The two rules whose test sets are most internally redundant are once again the two that look easiest and that carry the provenance asymmetry of Section 3.6 – three separate reasons to distrust a high score obtained that way, all pointing at the same two rules.

In Arm A the correction is negligible, ratio 1.04, and this is a point in the curated collection's favour that we want to state as plainly as we have stated its duplicates. At 40% identity its test partitions are essentially sets of singletons (1.0–1.1 sequences per cluster), so a sequence-level interval is already adequate there and the homology-aware partitioning did what it was meant to do within the test set. Not every diagnostic in this paper finds a problem.

What the small tasks cannot do, however, is identify a winner. In all five, the best pipeline's 95% cluster interval overlaps that of the *median* pipeline of 24 – for example on tumour T-cell antigen the best scores MCC 0.771 $[0.667, 0.867]$ against the median's 0.630 $[0.503, 0.754]$, and on anticancer 0.327 $[0.220, 0.427]$ against 0.267 $[0.155, 0.371]$. The overlap holds under the sequence bootstrap too, so it is not an artefact of the cluster correction: these

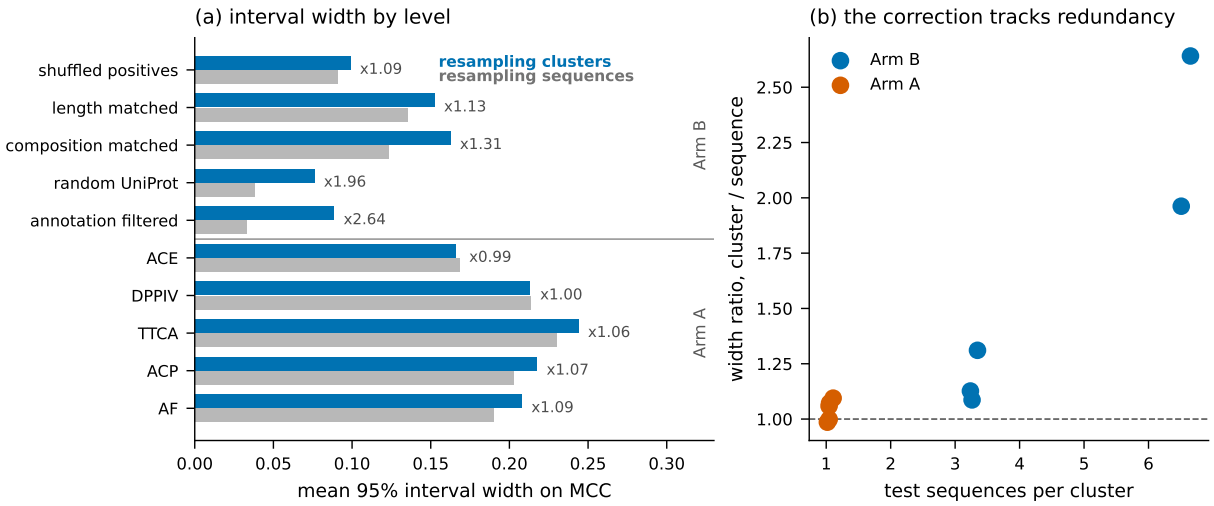


Figure 12: (a) Mean 95% interval width on MCC, resampling test sequences against resampling whole similarity clusters, with the ratio beside each pair. (b) The size of the correction is explained by one quantity: how many test sequences share a cluster. Arm B's proteome-derived rules carry 6.5–6.7 sequences per cluster and need the largest correction; Arm A's native partitions are near-singletons and need essentially none.

test sets are simply too small to separate the top of the leaderboard from its middle. That is an independent route to the same conclusion as Section 5.2 – winner identity is the fragile quantity – reached without any rank statistic, and it explains why rank transfer across small tasks is poor whatever the protocol.

5.5 Generalising over levels, not just over pipelines

Every interval so far resamples pipelines and therefore holds the five rules and ten tasks fixed (Section 2.7). Our headline claims are about protocols and tasks, so Table 10 recomputes them with the delete-one-level jackknife. The intervals roughly double in width, and for the flip rate they become close to uninformative.

Table 10: Delete-one-level jackknife intervals, which generalise over the *level* population, beside the pipeline-resampling intervals reported earlier. With four to ten levels these are necessarily wide; that width is the honest precision of a claim about protocols or tasks in general.

		mean τ_b, 95% CI		top-1 flip rate, 95% CI	
Label	levels	resampling pipelines	**level jackknife**	resampling pairs	**level jackknife**
B1	5 rules	[0.609, 0.818]	[0.561, 0.893]	[0.400, 0.900]	[0.060, 1.000]
B2	5 rules	[0.586, 0.769]	[0.465, 0.922]	[0.400, 0.900]	[0.580, 1.000]
B3	4 thresholds	—	[0.824, 0.954]	—	[0.000, 1.000]
A1	10 tasks	[0.351, 0.564]	[0.395, 0.563]	[0.800, 0.956]	[0.853, 1.000]

Three consequences, and we state the uncomfortable one first.

The winner-flip rate does not support a population claim in Arm B. At five rules the level-jackknife interval on B1's flip rate is $[0.060, 1.000]$. The observation that most rule pairs crown a different pipeline is solid *for these five rules*, and it is the experience a practitioner has when switching between two published negative-set definitions; but five levels cannot locate that rate in a wider population. In Arm A, with ten tasks, the interval is $[0.853, 1.000]$ and the claim does survive generalisation.

Arm A's agreement is the better-determined of the two, because ten levels beat five: its two intervals nearly coincide ($[0.351, 0.564]$ against $[0.395, 0.563]$), whereas B2's widens from

$[0.586, 0.769]$ to $[0.465, 0.922]$.

The two-arm difference survives under the primary analysis and not under the post hoc one. Treating the rule set and the task set as independent samples of levels, the jackknife standard error of the difference is $\sqrt{\mathrm{SE}_B^2 + \mathrm{SE}_A^2}$:

Arm B splitter	**B1** pre-registered *(primary)*	B2 balance-preserving *(post hoc)*
difference in τ_b	+0.218	+0.185
level-jackknife SE	0.097	0.126
95% CI, generalising over levels	[+0.029, +0.408]	[−0.061, +0.431]
two-sided p	**0.024**	0.141
95% CI, resampling pipelines only	[+0.117, +0.317]	[+0.110, +0.269]

Under B1, the pre-registered and primary comparison, task variation disturbs the ranking significantly more than protocol variation even when the inference is generalised over levels ($p = 0.024$). Under B2 it does not ($p = 0.141$), because correcting the class-balance artefact lowers Arm B's agreement toward Arm A's and simultaneously widens its level-jackknife interval. **The claim therefore stands on the primary analysis and is weakened, not strengthened, by the post hoc correction** – the opposite of the direction a reader might suspect a post hoc analysis of being chosen for, and we flag it rather than reporting only the favourable pairing. Because the sign agrees in both and the primary is significant, we retain the claim, restricted to these five rules and ten tasks.

5.6 Design-simulation limits, pre-specified

Before any data were analysed we characterised the estimator. The mean of pairwise τ_b has null SD 0.069 at 12 pipelines and 5 protocols, against 0.219 for a single pair; minimum detectable disagreement at 80% power is $\tau_b \leq 0.646$ at 16–20 pipelines, which is why a pre-registered observation of 0.727 was reported as inconclusive rather than as a near-miss. The simulation assumed a ceiling of 0.955 at 5% run noise; the *measured* ceilings were 0.867 (Arm B) and 0.838 (Arm A), so real run-to-run noise is larger than assumed and the simulation was, if anything, optimistic (Supplementary Figure S2).

6 Discussion

6.1 What this study can and cannot settle

Before any interpretation, the boundary of the evidence, because every claim below sits inside it and the reader should not have to reach the limitations section to find out.

Arm B varies five negative-set construction rules around one fixed positive set of 5,187 peptides. The positive set is ours to the extent that we chose which published collection to build on; the five rules span the families behind eleven published definitions but are not those eleven implementations. Conclusions attach to these five rules on this corpus.

Arm A varies ten biological tasks that come from a single collection, curated by one group with one homology-aware partitioning pipeline [2]. This is the limitation that most constrains the paper's reach. Those ten datasets share a curator, a label convention, a negative-set philosophy and a splitting procedure. What varies between them is the biological question; what does *not* vary is everything about how the benchmark was built. Arm A therefore measures **task transfer under a fixed protocol**, and it is not a cross-benchmark comparison in any form.

We did not compare independently constructed benchmarks. That was the original motivation and it remains open (Section 2.6, deviation 1). Nothing here is evidence that independently built benchmarks agree or disagree, and a reader who wants that question answered

should treat this paper as instrumentation and a set of measured artefacts rather than as the answer. Settling it needs several genuinely independent resources harmonised to a common task definition and each reproduced to its own published baseline – the gate we pre-specified and could not apply.

The honest size of the contribution, then: a decomposition showing which design choice owns the reported number, four measured failures of a splitting method the field has standardised on, two agreement statistics reported against measured ceilings instead of nominal maxima, and a pre-registered null. All of it demonstrated on one corpus and one curated task collection.

6.2 What we set out to test, and what we found instead

We pre-registered the hypothesis that leakage-controlled evaluation protocols would rank a fixed set of methods differently, and fixed the decision thresholds in advance. Under the pre-registered analysis **that hypothesis was not supported**: normalised agreement was 0.881 against a threshold of 0.85, and $\tau_b = 0.727$ fell inside the band we had declared inconclusive for this design. Had we not fixed those thresholds in advance, 0.727 would have been straightforward to present as disagreement.

Because the result was a null, the interesting question became why the intuition was wrong, and in what narrower sense it is right.

6.3 The ordering is stable; the winner is not

Two facts must be held together: mean τ_b across protocols is high, and 9 of 10 protocol pairs name a different best pipeline. These are not in tension. τ_b weights all pairwise orderings equally, and most are stable – the weakest pipelines occupy the bottom places under every protocol, contributing concordant pairs regardless. The reordering concentrates among the leading pipelines, which are close in score and therefore easy to permute. **A rank correlation computed over a whole leaderboard is the wrong instrument for a field that reads only the top of it.**

That the leading places are close enough to permute easily has a consequence we did not anticipate and that cuts against our own framing. If the top pipelines are separated by less than run-to-run noise, then the seed alone should move the winner – and it does, in 54.5% of comparisons (Section 5.2). Protocol choice flips the winner more often than that, in 76.0% of comparisons, and the excess appears under both splitters – though with five rules we report it as a measurement on those rules rather than as an inference about protocols, since resampling the rules leaves the rate undetermined. But the honest summary is two-part: **the winner is a noisy statistic before any protocol is varied, and varying the protocol makes it substantially noisier**. The practical recommendation that follows is not only to report rankings under a second negative-set rule, but to report the winner's stability across seeds at all – which almost no benchmark currently does, and which costs five reruns.

6.4 Protocol moves the score; interaction moves the rank

The negative-set rule accounts for 71.0% of the variance in MCC, against 14.8% for the pipeline and 0.6% for the clustering threshold. The single largest determinant of a reported number in this literature is a construction choice that has no standard and is often described in one sentence.

Protocol main effects contribute *nothing* to rank variance, but we want to be exact about why, because we initially presented this as a finding and it is not one. Ranks are computed within each protocol cell, so every cell's mean rank is $(K + 1)/2$ by construction and any cell-defining factor has zero between-group sum of squares identically, for any data (Section 3.4). The useful content of the rank decomposition is in the interactions, which are free to vary: pipeline × rule is 8.87% $[8.30, 9.43]$, seven times pipeline × threshold at 1.26% $[0.97, 1.55]$.

We still recommend decomposing both responses, with the caveat made explicit: when a paper claims a design factor matters, ask whether it matters to the number or to the conclusion drawn from the number – but read a zero on the rank response as a property of the estimator unless the factor is nested rather than crossed with the ranking unit.

6.5 Task transfer is the larger problem

Rank agreement is lower across biological tasks than across evaluation protocols. Generalising over levels, which is the population the claim is about, the difference is $+0.218$ $[+0.029, +0.408]$, $p = 0.024$ under the pre-registered splitter and $+0.185$ $[-0.061, +0.431]$, $p = 0.141$ under the post hoc one; holding the levels fixed and resampling only pipelines gives $+0.218$ $[+0.117, +0.317]$, $p = 0.001$ and $+0.185$ $[+0.110, +0.269]$, $p < 0.001$ respectively. The claim rests on the pre-registered splitter, and on these ten tasks and five rules (Section 6.1).

Seven different pipelines win across ten tasks and one moves 20 of 24 positions. The field has invested heavily in standardising the splitting protocol – the factor our data identify as the *smaller* of the two – and has no convention at all for either the negative-set rule or for reporting whether a method's advantage transfers beyond the task it was demonstrated on.

6.6 Four artefacts of a method the field has standardised on

Cluster-disjoint splitting does not deliver what it is assumed to. It is not pairwise-disjoint, and residual leakage *rises* with nominal strictness, reaching 21.6% overall and 39% for whole-protein negatives. It destroys class balance, up to 99.5% positive on a 1:1 corpus. It induces covariate shift that in two tasks reverses the sign of a discriminative feature. And the diagnostic a benchmark would naturally report – the cluster-to-sequence ratio – moves reassuringly in the opposite direction to all of these.

None of this is an argument against homology-aware splitting; random splitting is plainly worse. It is an argument that the residual is large, systematic, largest where the claim sounds strongest, and currently unreported.

6.7 What is confirmatory, what is post hoc, and what is an observation

The strongest sentences in this paper are not the best-supported ones, and a reader is entitled to see that sorted out explicitly rather than inferred from hedging. Only one result is confirmatory, and it is a null.

The last row deserves naming for what it is. Statements about what "the field" has standardised, and about what benchmark results mean in general, are **interpretation**: they are our reading of why the measured pattern matters, not findings. We have kept them because we think they are the right reading and because a paper that refuses to interpret its own results is less useful, not more rigorous. But they are not evidence, they are not pre-specified, and nothing in Table 11 above them depends on accepting them.

6.8 Limitations

- **Scope: five rules on one corpus, ten tasks from one curator.** Stated in full in Section 6.1 and not restated here. In short: Arm B's corpus is ours and is instrumentation rather than a proposed standard; Arm A varies the biological question at a fixed protocol; the cross-benchmark comparison was not performed and remains open.
- **The balanced splitter is post hoc.** It was built after the pre-registered null. We report it as a sensitivity analysis and do not claim it confirms the hypothesis.
- **The RBF SVM is approximated** by a Nyström map – the one substitution that changes a model rather than its scheduling. We validated it against the exact kernel on 15 matched

Table 11: Evidential status of every claim we make. *Confirmatory* means tested against a threshold fixed before the data were analysed. *Post hoc* means the analysis was designed after seeing results. *Descriptive* means a measurement on these specific corpora with no inferential claim attached, which is where our rhetorically strongest observations sit.

Claim	Status	Generalises to
H1: protocols rank methods differently (**B1**, normalised 0.881 vs threshold 0.85) – **not supported**	confirmatory	these 5 rules
Task variation disturbs ranking more than protocol variation (+0.218, $p = 0.024$ over levels)	secondary, pre-specified estimator	these 5 rules, 10 tasks
Negative-set rule owns 71.0% of score variance vs pipeline 14.8% (H3)	secondary, pre-specified	this corpus
Seed component negligible (H4)	secondary, pre-specified	this corpus
Winner flips above its seed-noise ceiling, *across tasks* (0.933 vs 0.570; level-based CI $[0.853, 1.000]$)	post hoc (ceiling added after an audit of our own reasoning)	these 10 tasks
Winner flips above its seed-noise ceiling, *across negative-set rules* (0.760 vs 0.545; level-based CI $[0.060, 1.000]$)	post hoc, **descriptive only** – five levels cannot locate the rate	these 5 rules
Balance-preserving splitter lowers agreement to 0.800 (**B2**)	post hoc	this corpus
Residual pairwise leakage rises as the threshold tightens (1.4% $\rightarrow$ 21.6%)	descriptive	this corpus
Cluster-disjoint splitting destroys class balance (to 99.5% positive)	descriptive	this corpus
Covariate shift inverts a feature's sign in 2 tasks	descriptive	those 2 tasks
11 of 16 released datasets contain cross-split duplicates	descriptive, verifiable	that collection
Two-feature baseline reaches MCC 0.79 on proteome negatives	descriptive	this corpus
"The field has standardised the factor that matters less to the ranking"	interpretation	–

cells (Section 2.10): it costs about 0.03–0.04 of AUROC and MCC but preserves the rule-level ordering ($r = 0.997$ on MCC). The approximation lowers one pipeline's absolute scores; it does not move the rankings this paper reports.

- **CPU-only hardware excluded two pipeline families** – end-to-end convolutional and recurrent models, and a 3B-parameter language model – so we cannot say whether end-to-end sequence models are more or less protocol-sensitive.
- **Covariate-shift correlations are marginal** at ten tasks ($p = 0.054$, $p = 0.090$) and are reported as suggestive.
- **H1 was pre-specified, not registered.** The analysis plan is dated and released but was not externally deposited, and its primary hypothesis originally named five published benchmarks as the comparison unit rather than the protocol levels we ultimately varied (Section 2.6, deviation 1). A reader should discount H1's confirmatory status accordingly.
- **The rank-variance decomposition cannot speak to protocol main effects.** They are zero by construction, not by measurement (Section 3.4). Our evidence that protocol preserves ordering is the rank-agreement and flip-rate analyses, not the decomposition.
- **Class provenance is asymmetric, and we did not test pretraining exposure.** Under the two proteome-derived rules every negative and 7.9% of positives are verbatim reviewed-UniProt entries (Section 3.6). That is an overlap audit against the *current* release; ESM-2 was trained on UniRef from release 2021_04, so our audit bounds exposure in neither direction, and we do not claim it does. Whether the encoder had seen these sequences remains untested.
- **The negative realisation was fixed per rule in the main grid.** A single draw per rule was embedded, because embedding five draws at four ESM-2 scales was not affordable on CPU, so the reported seed component varies the partition and model initialisation but *not* which negatives were sampled. We quantify draw-to-draw variability separately on the embedding-free pipelines (Section 3.9): the rule dominates the draw about two to one, but for the two matched-negative rules the draw alone moves the ranking to normalised 0.65–0.69, so their variance components are conditional on the realisation rather than marginal over it.

6.9 Recommendations

In order of value per unit of effort.

1. **Measure and publish residual pairwise leakage.** Search the test partition against the training partition and report the fraction of test sequences with an above-threshold hit. Cluster-disjointness does not imply it, and it costs one search.
2. **Report test-set class balance per split.** It is one number and it can be 0.99 without anyone noticing.
3. **Rerun with five seeds and report whether the winner survives.** In our grids re-seeding alone changes the top-ranked pipeline in 54.5% of comparisons (Section 5.2). A leaderboard that names a best method from one run is reporting a statistic it has not shown to be stable, and the check costs five fits.
4. **Include a trivial baseline.** Length and net charge cost nothing. If two features place near the top, fix the negative set before publishing the model.
5. **Report rankings under at least two negative-set rules**, and state whether a method's advantage transfers to a second task.
6. **Quote agreement against a measured ceiling, not against** 1.0**.** Seed replicates give the ceiling free; ours were 0.838 and 0.867.

7. **Report the metric you chose and the ranking under alternatives.**
8. **Assign clusters in random order, not by size.** Size-ordered assignment is what produces both the class-balance collapse and the worst residual leakage; a uniform shuffle of cluster order cut mean leakage from 0.107 to 0.055 and balance error from 0.305 to 0.009 in our stress test, and costs one line (Table 2). If leakage matters more than balance, order by how little each cluster bridges to the rest.
9. **Publish split files, not split procedures**, and state the clustering tool, threshold *and* coverage parameter. For matched-negative rules, publish the *realisation* too: redrawing the negatives under a fixed rule moved our rankings to normalised 0.65–0.69 (Section 3.9), so the rule alone does not specify the benchmark.
10. **Check for exact duplicates across your own split.** Eleven of sixteen carefully built datasets contained them.

6.10 Conclusion

Homology-aware splitting corrected a real problem and the field is better for it. It did not make evaluation interchangeable across the levels we varied, but the failure is more specific than expected and more actionable for it. Protocol choice does not scramble the ranking of methods – the broad ordering is largely preserved, retaining 88% of the agreement that rerunning the same protocol achieves – while it changes which method is crowned in 9 of 10 rule pairs, and it changes the apparent difficulty of the task by 0.6 MCC. Varying the biological question disturbs the ranking more still. Since the literature advances by adopting whatever placed first, those are the failures that matter.

All of it holds for **five negative-set rules on one corpus** and **ten tasks from one curator**, the boundary set out in Section 6.1. Within those levels the effects are large and measured; generalised beyond them our own intervals are wide enough that we decline to make the claim.

The splitting method itself needs attention independently: as standardly implemented it leaves up to 21.6% of the test set leaking at its strictest setting, can return a test partition that is 99.5% one class, and can invert the sign of the very feature a model depends on – while the statistic a benchmark would report to reassure a reader moves in the opposite direction to all three.

Data and code availability

All analysis code, the frozen pipeline specification, the protocol configuration, the complete record of all model fits and the scripts that regenerate every figure and table are released as a single archive, `bra-harness.tar.gz`, with SHA-256
`e43629198237357d4412a2a28740da0829c4cdeaaae34263848889ffad608f35`
Deposit URL/DOI: INSERT_AT_DEPOSIT.

The dated analysis plan, including all three revisions and the complete deviations log discussed in Section 2.6, is released in the archive as `preregistration.md`. It was not deposited with an external registry and carries no third-party timestamp; we say so there as well as here.

`reproduce.sh` reruns the study end to end from a clean checkout, including input retrieval, and is idempotent and resumable at both the embedding stage (cache keyed by sequence hash) and the grid stage (one file per cell).

The Arm B corpus is not redistributed: its positives derive from DBAASP via the AMP-Benchmark positive set and its negatives from UniProt, and those terms govern redistribution. We release the construction scripts with SHA-256 hashes of both inputs, which reconstructs

the corpus given the same source files. The Arm A datasets are redistributed by their original authors and are not duplicated here.

What the released run records cover. "4,400 run records" refers to the two main factorial designs and nothing else. The supplementary analyses add 530 further model fits and are deposited in the same archive, each with the script that produced it and the table it writes.

Exact inputs needed to rebuild the corpora. Reproduction depends on pinned inputs, not just on pinned code, so we state each one and how it is verified:

- **Arm B positives** – the published positive set of Sidorczuk et al. [1] (AMPBenchmark), SHA-256 recorded in the archive. 5,187 sequences after removing the three homopolymers named in the Methods.
- **Arm B negatives** – reviewed UniProt (SwissProt), **retrieved 2026-10-05**, 575,749 records of which 573,063 are fully canonical. Because UniProt releases are not immutable at a stable URL, the archive records the SHA-256 of the retrieved FASTA; rebuilding against a later release will not reproduce the corpus byte-for-byte, and `reproduce.sh` fails loudly on a hash mismatch rather than proceeding with different negatives.
- **Arm A datasets** – the AutoPeptideML peptide benchmark collection [2], 16 datasets, used as released; per-dataset row counts and SHA-256 values are in the archive, and the sizes are printed in Supplementary Table S3 so a reader can check a download without running anything.
- **Tools** – MMseqs2 18-8cc5c+ds-1 and the Python versions listed above; the clustering call is given in full in the Methods, coverage parameter included.

Status of the deposit. The archive hash above is final and the deposit identifier must be inserted before submission. We flag this rather than leave it implicit: a reproducibility claim with no resolvable location is not a claim a reader can act on, so until the DOI is present the availability statement should be read as an undertaking rather than as a completed deposit.

Software: Python 3.12.15, numpy 2.5.3, pandas 3.0.6, scipy 1.18.1, scikit-learn 1.9.1, XGBoost 3.4.1, PyTorch 2.14.1 (CPU), MMseqs2 18-8cc5c+ds-1, qmap-benchmark 0.1.1. Reference hardware: AMD Ryzen 9 8940HX, 32 threads, no GPU.

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Conflicts of interest

None declared.

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